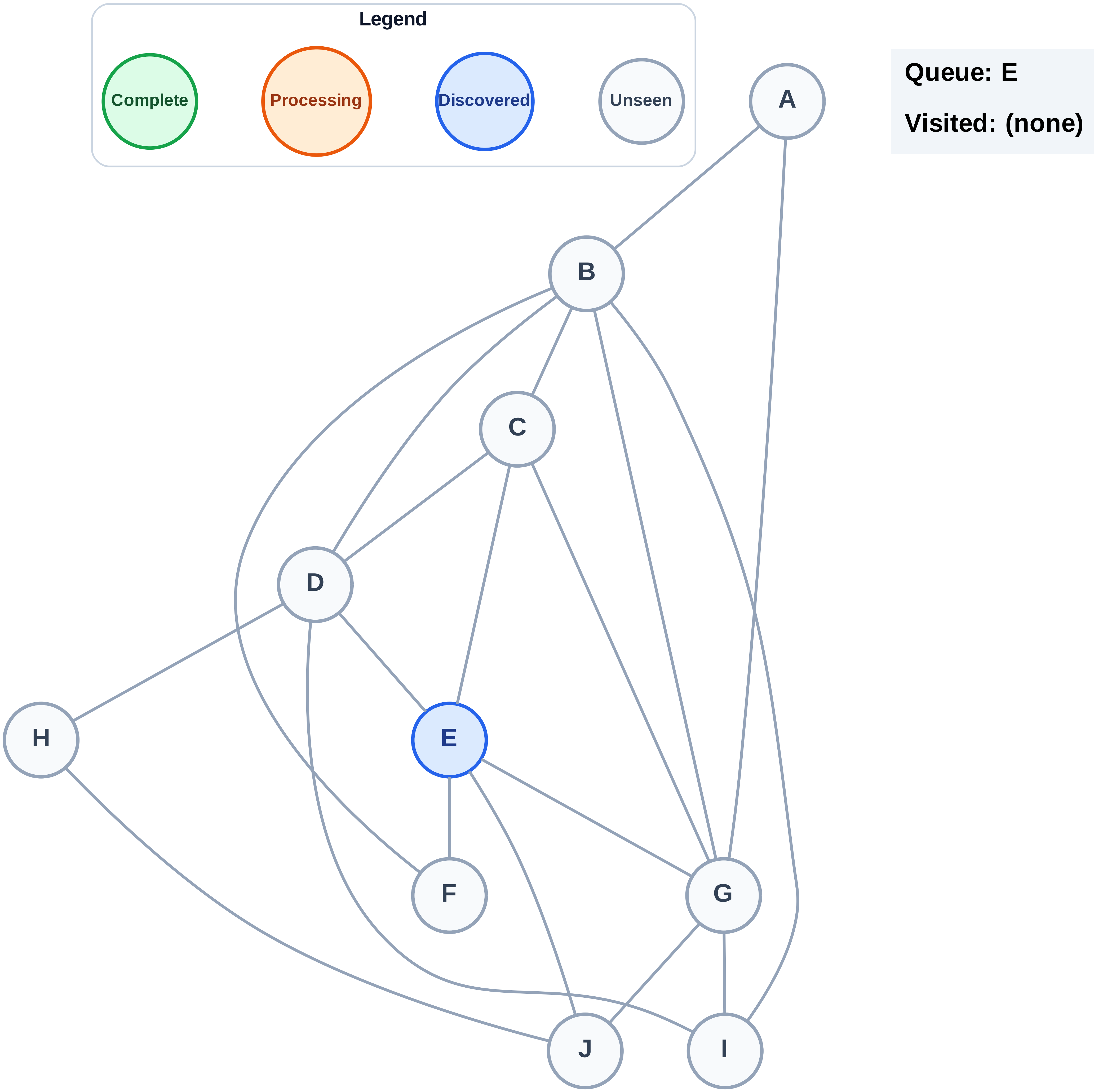
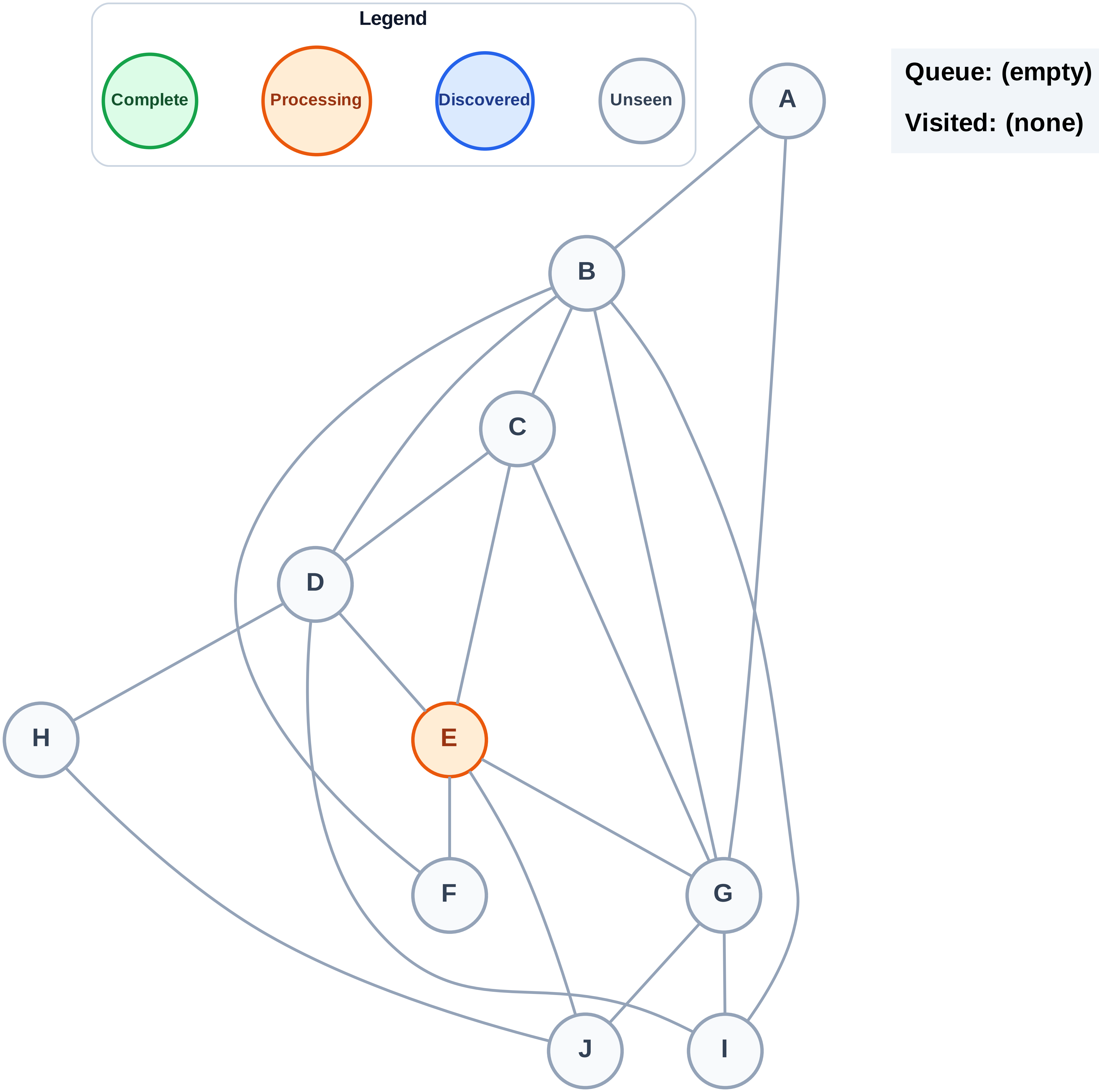
# BFS Traversal

Step 1: Start at E



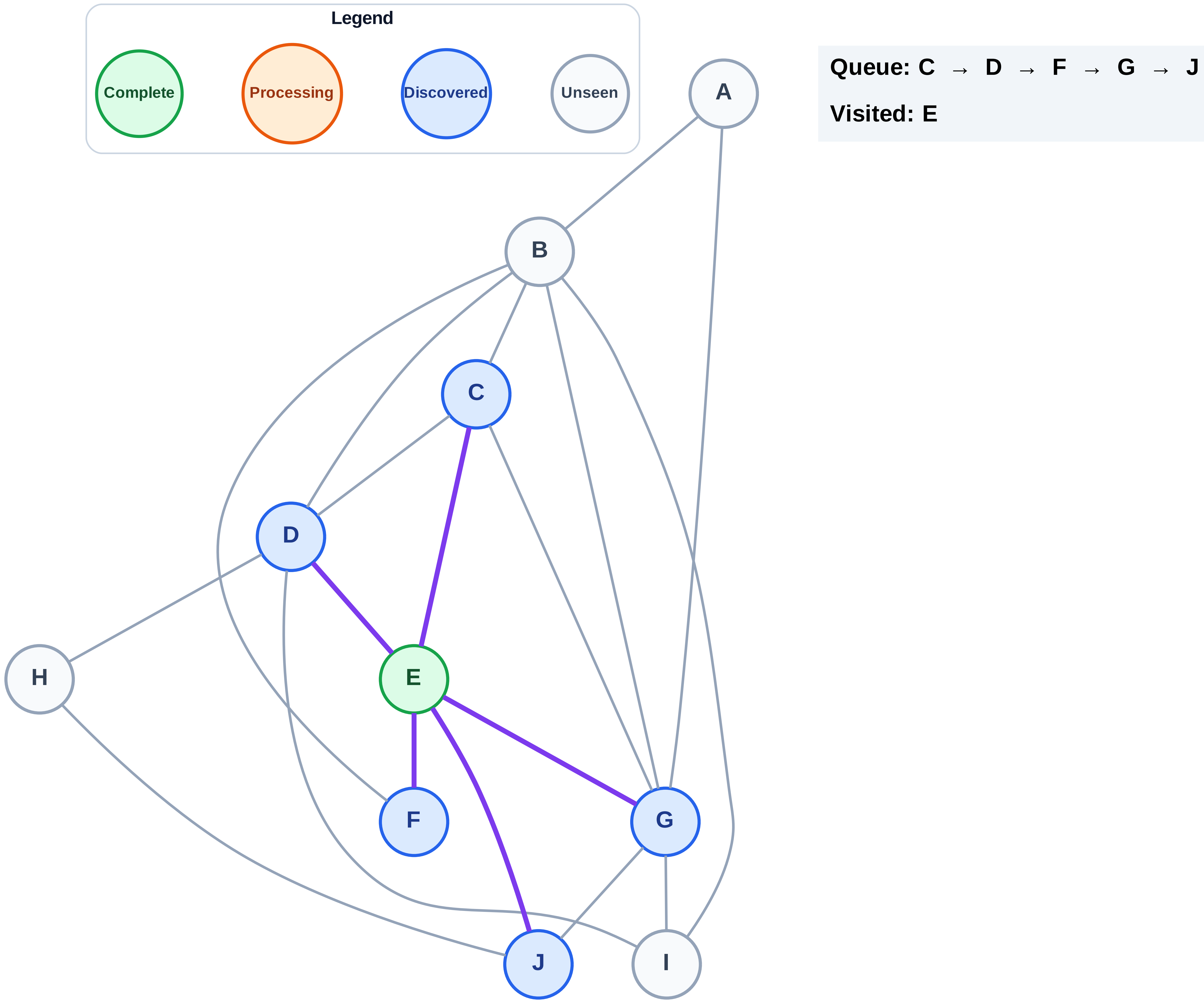
# BFS Traversal

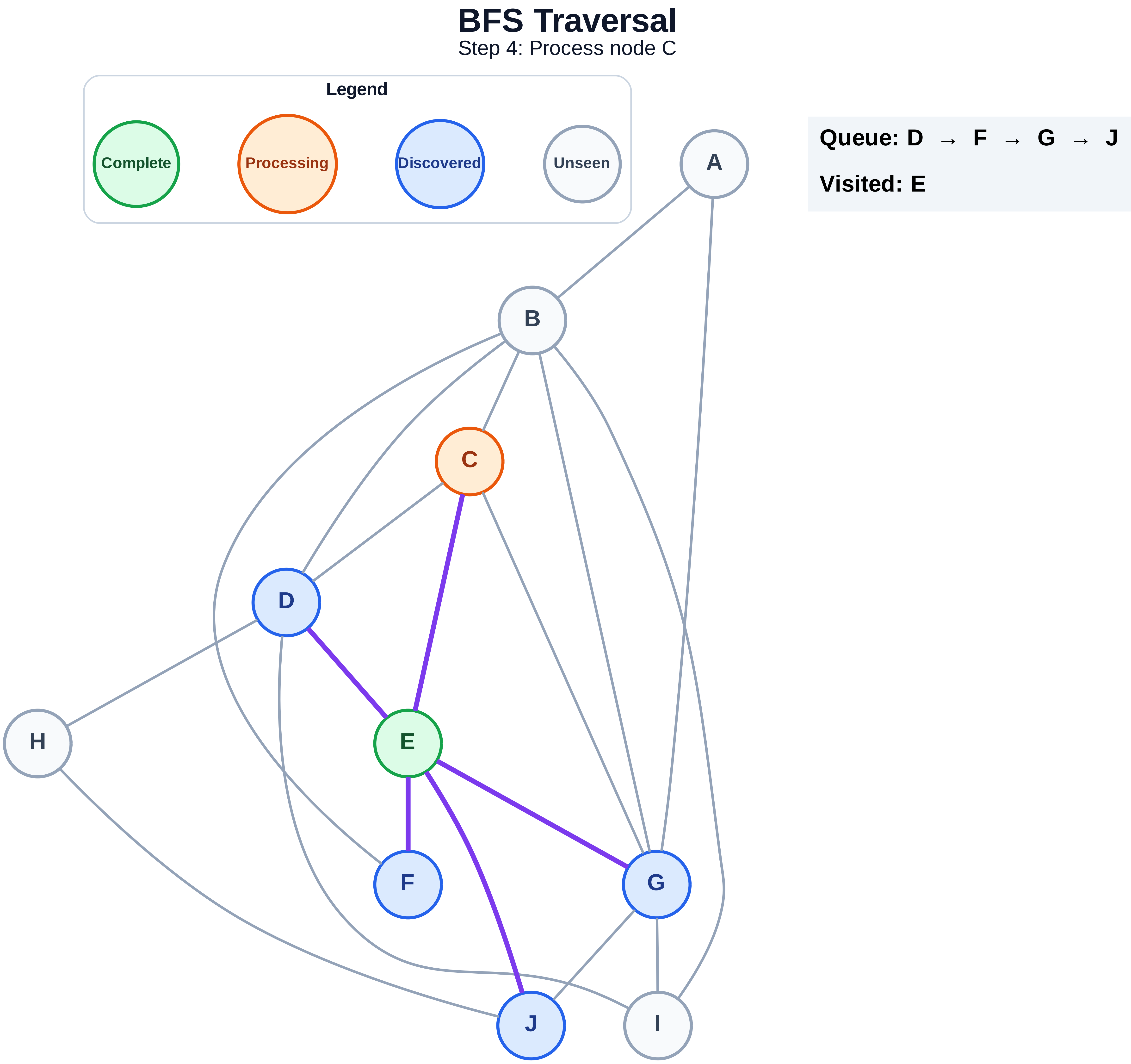
Step 2: Process node E



# BFS Traversal

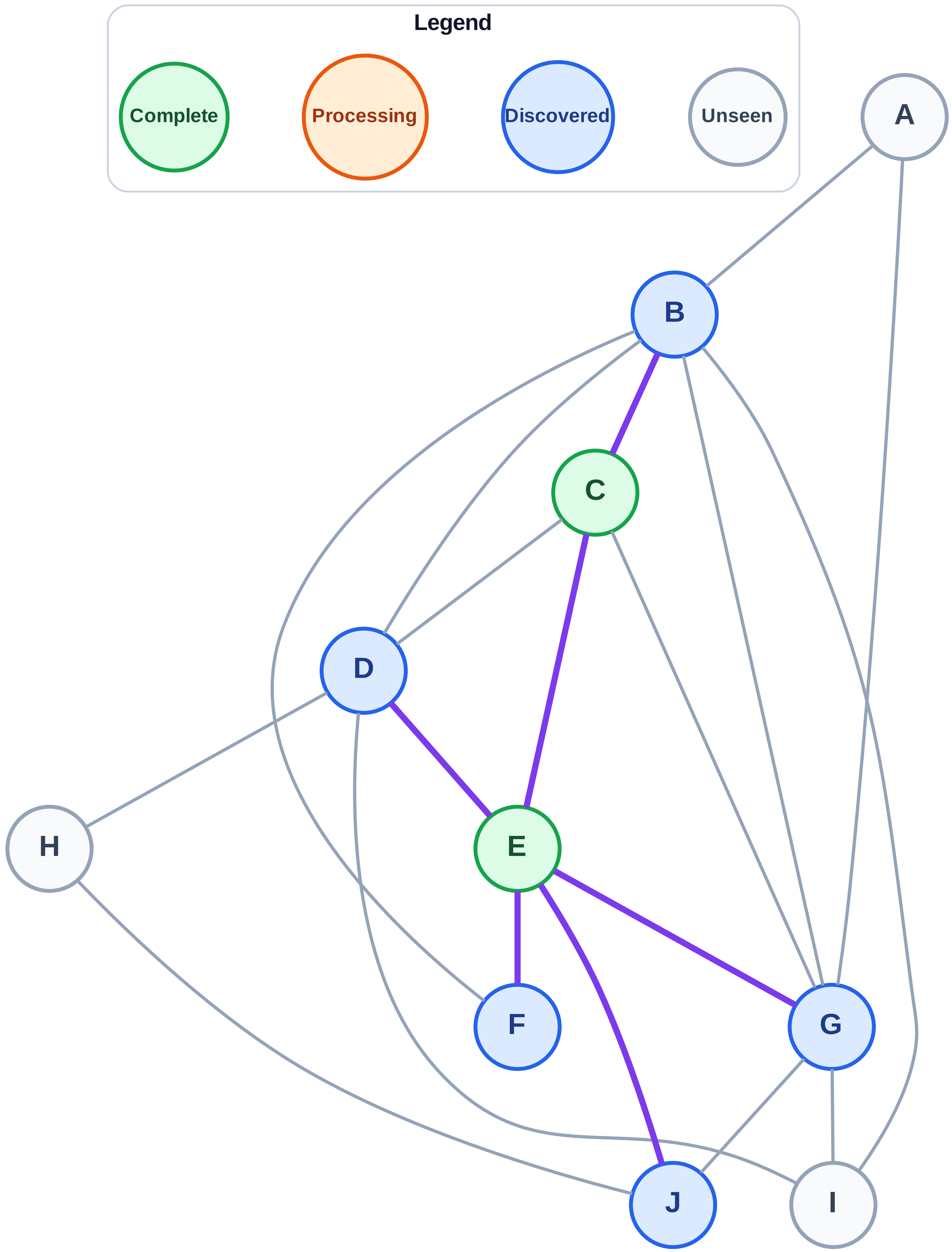
Step 3: Discover C, D, F, G, J from E




BFS Traversal
Step 4: Process node C
Legend
Complete
Processing
Discovered
Unseen
Queue: D → F → G → J
Visited: E
A
B
C
D
E
F
G
H
I
J

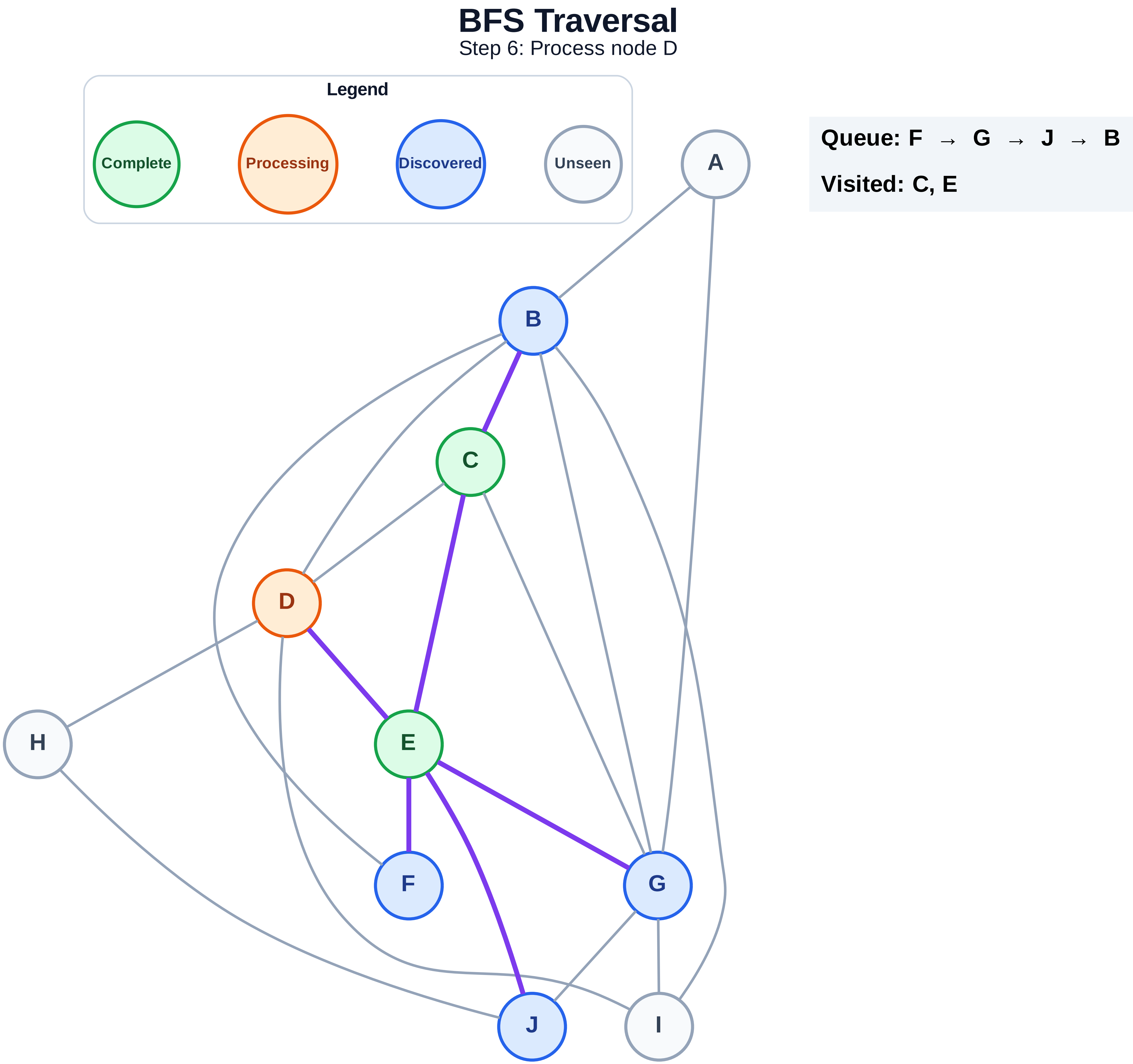
# BFS Traversal

Step 5: Discover B from C



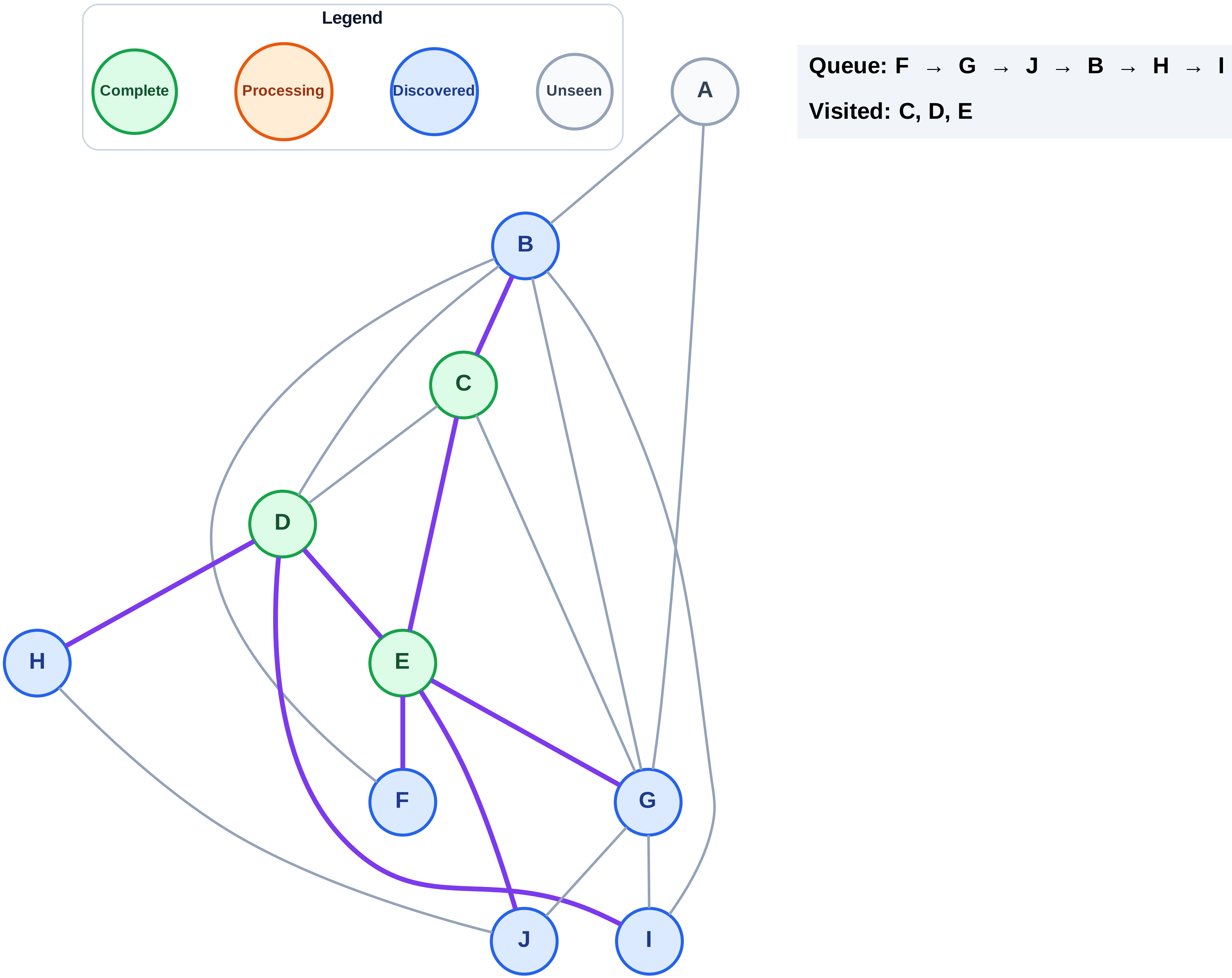

**Queue: D → F → G → J → B**

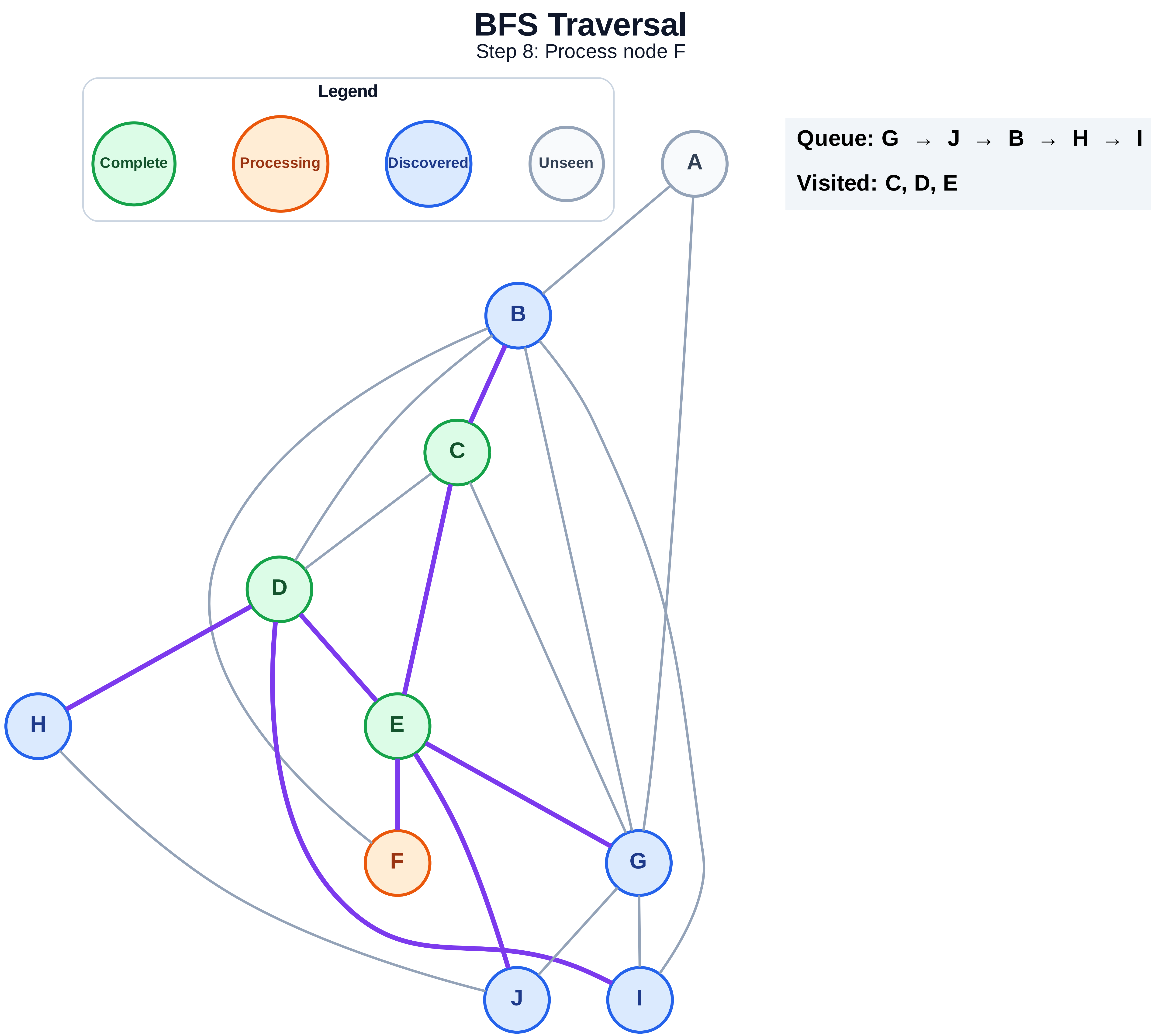
**Visited: C, E**


BFS Traversal
Step 6: Process node D
Legend
Complete
Processing
Discovered
Unseen
Queue: F → G → J → B
Visited: C, E
A
B
C
D
E
F
G
H
I
J

# BFS Traversal

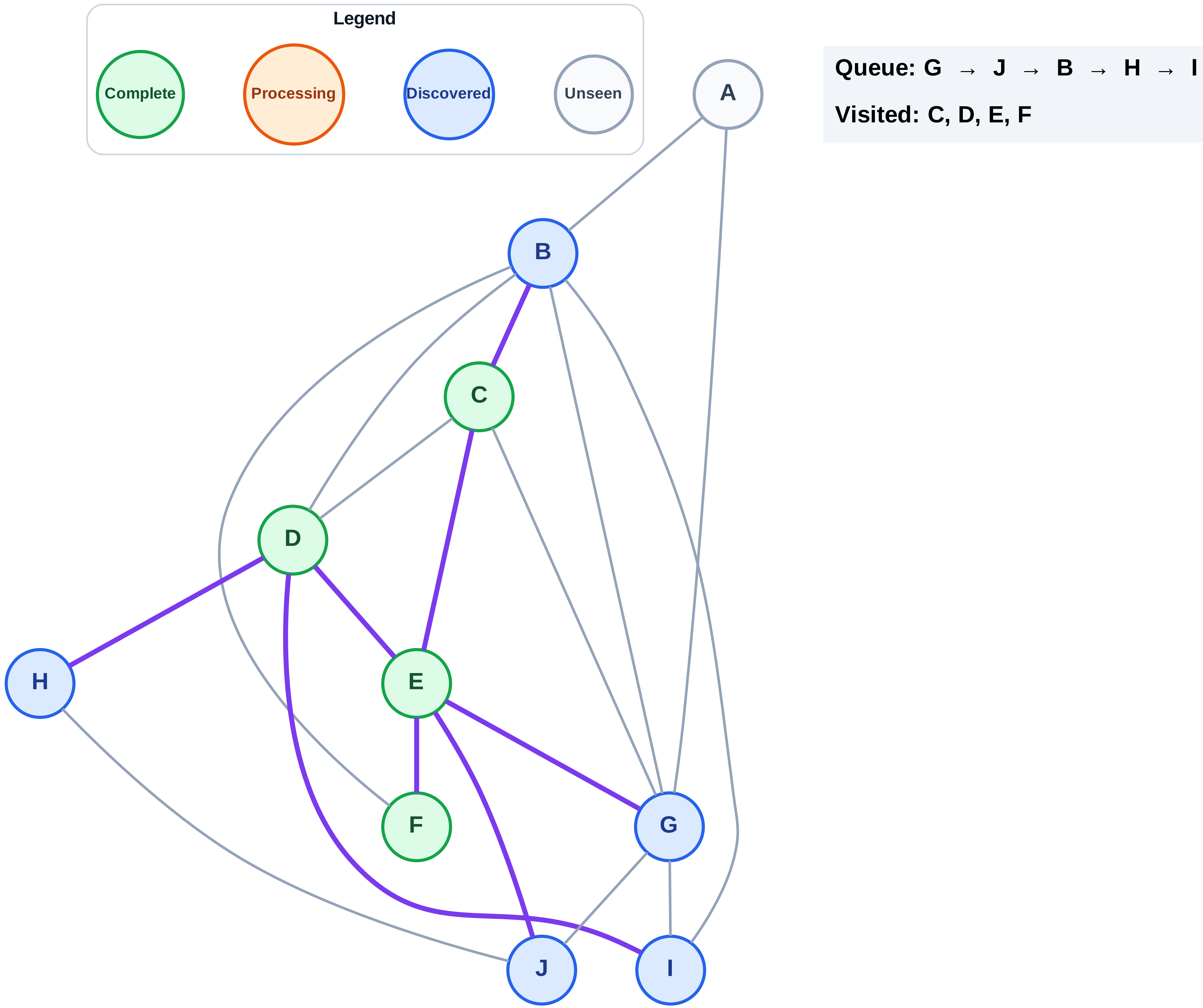
Step 7: Discover H, I from D




BFS Traversal
Step 8: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: G → J → B → H → I
Visited: C, D, E
A
B
C
D
E
F
G
H
I
J

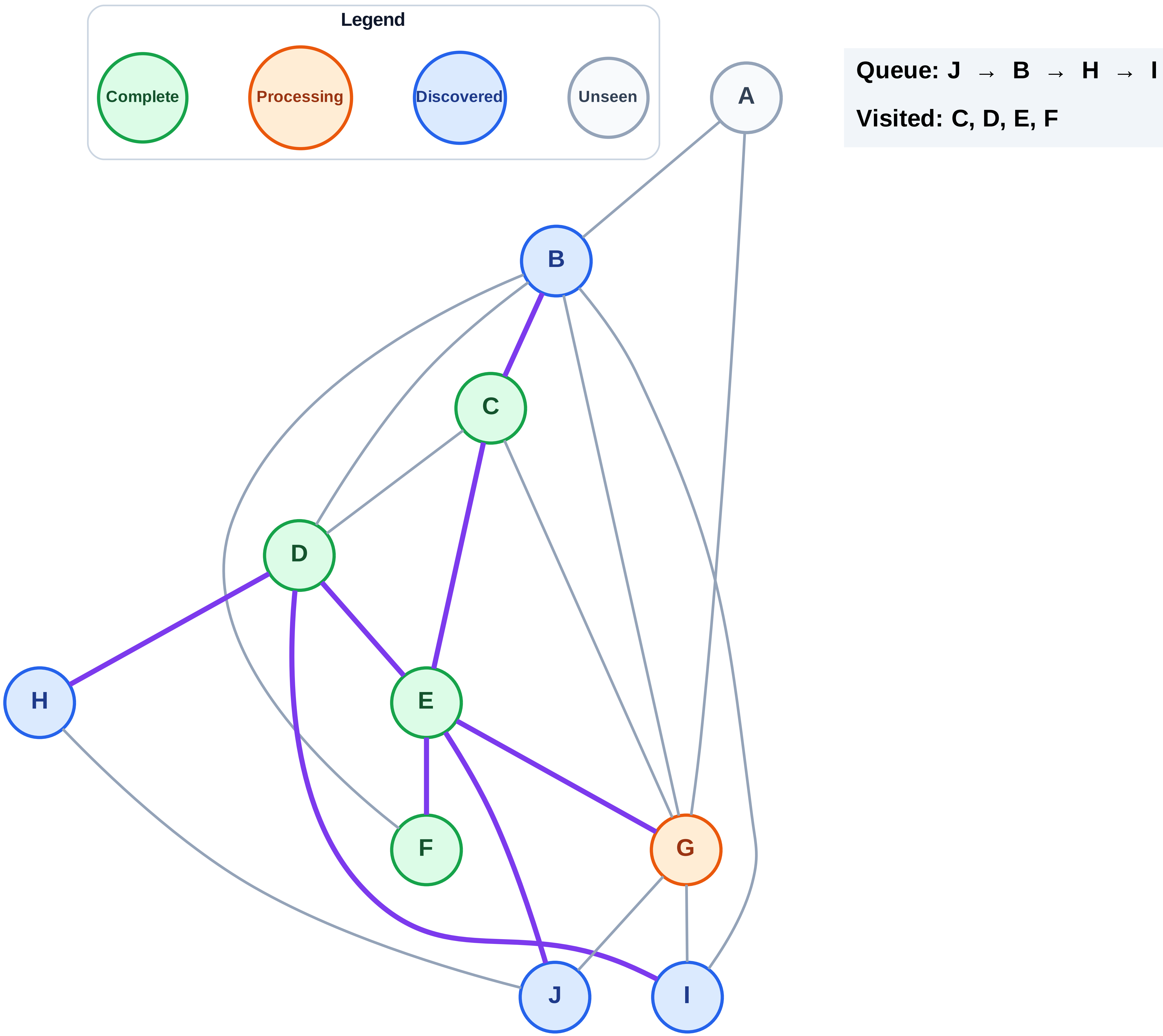
# BFS Traversal

Step 9: No new neighbors from F



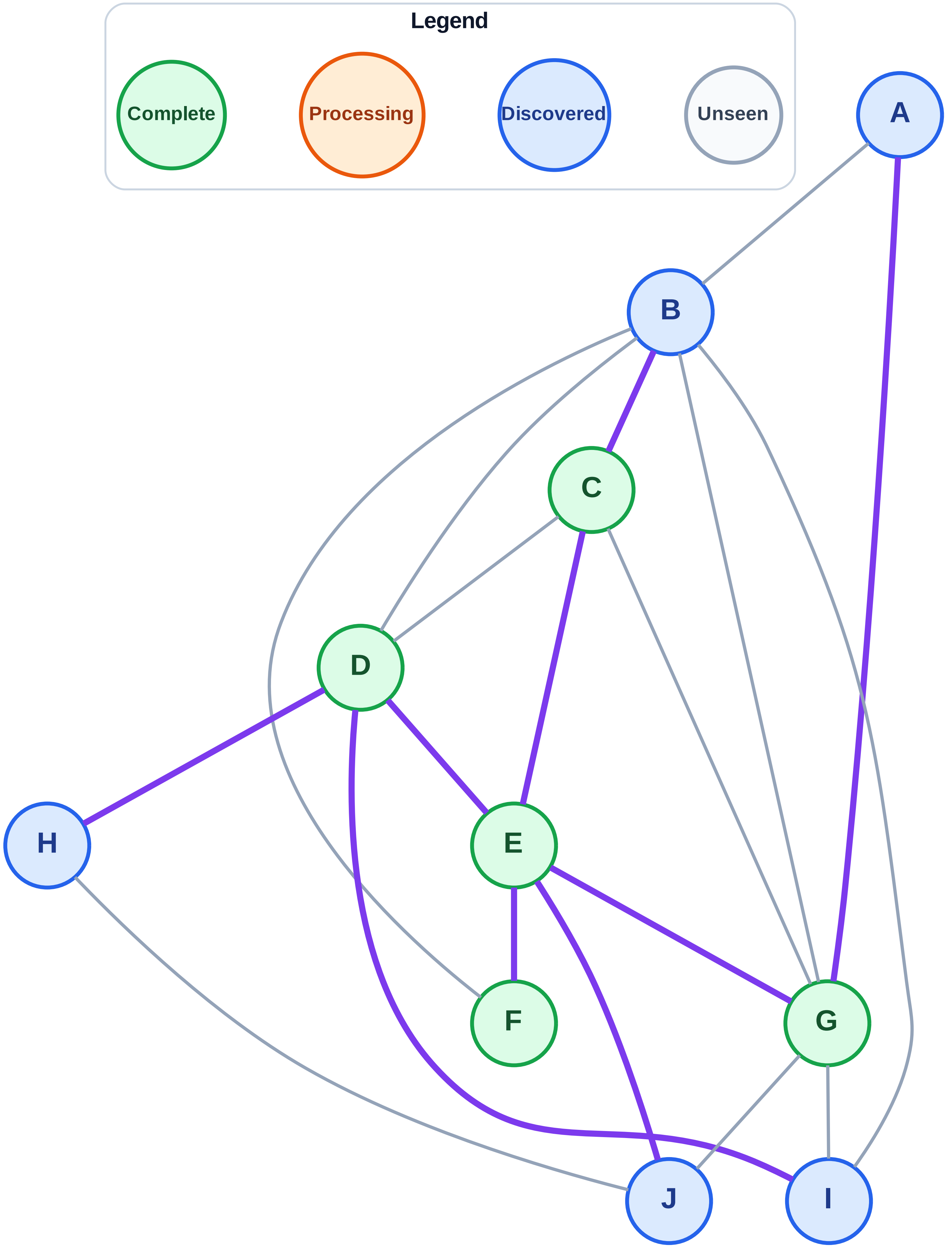
# BFS Traversal

Step 10: Process node G



# BFS Traversal

Step 11: Discover A from G

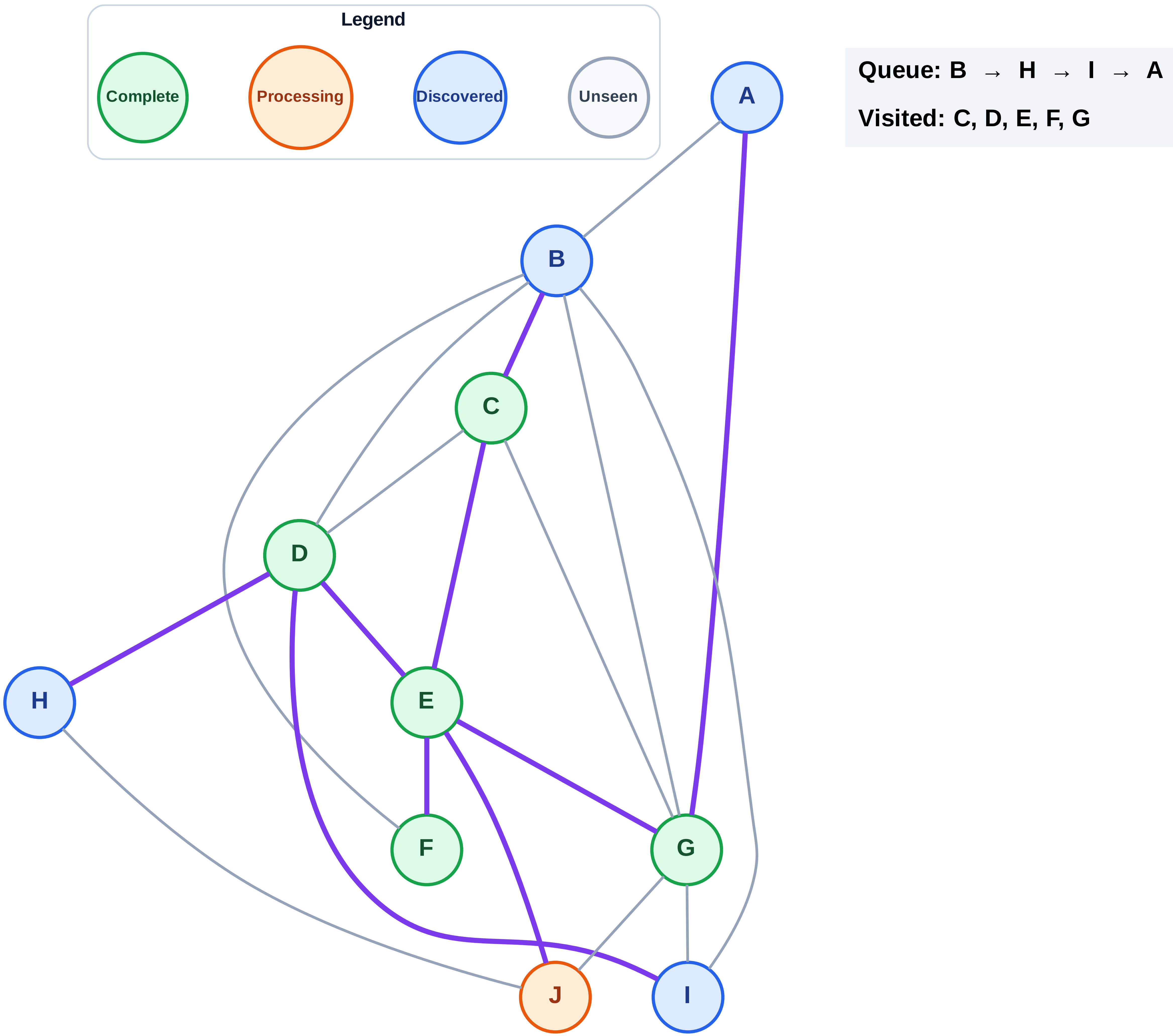



**Queue: J → B → H → I → A**

**Visited: C, D, E, F, G**

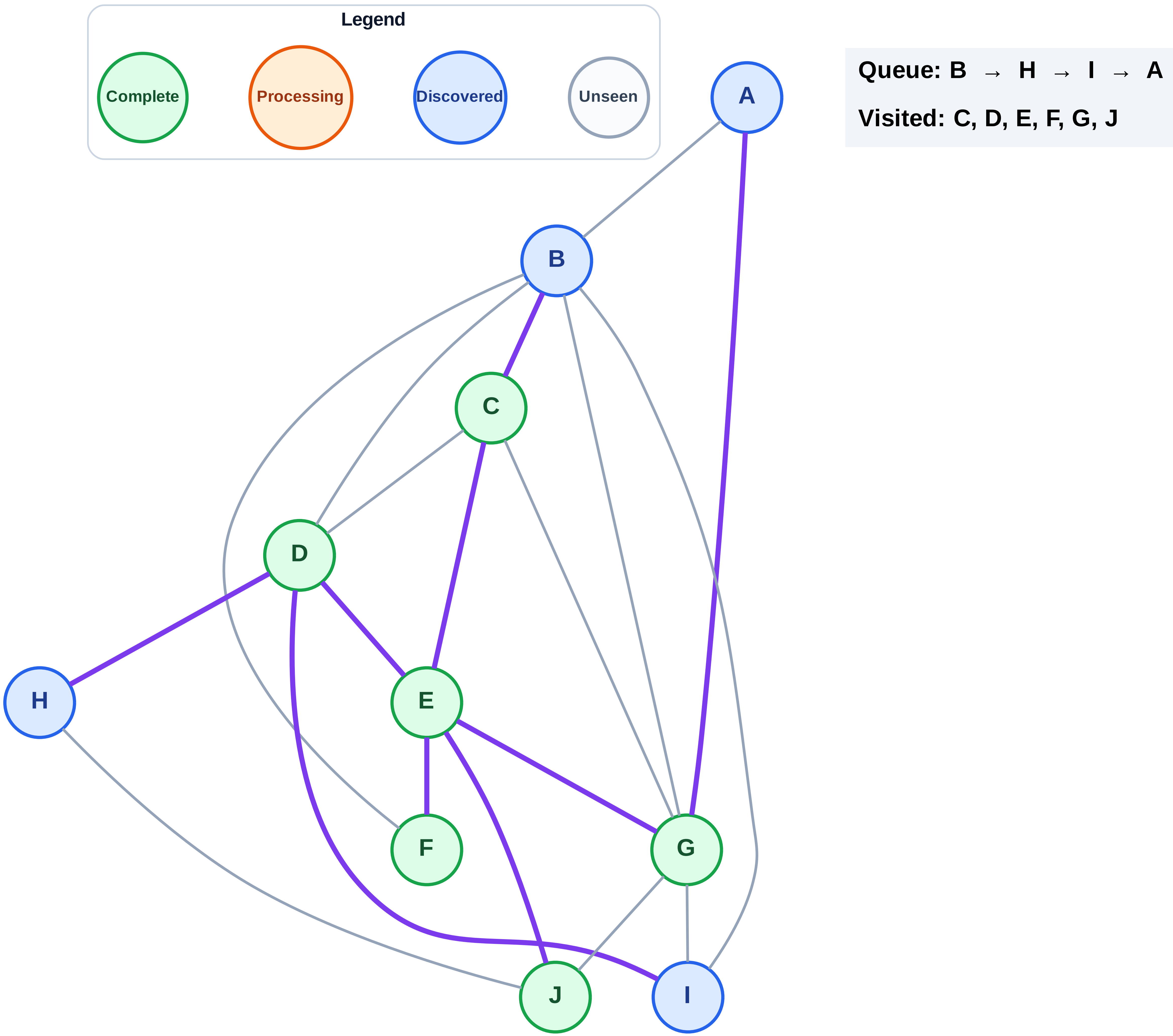
# BFS Traversal

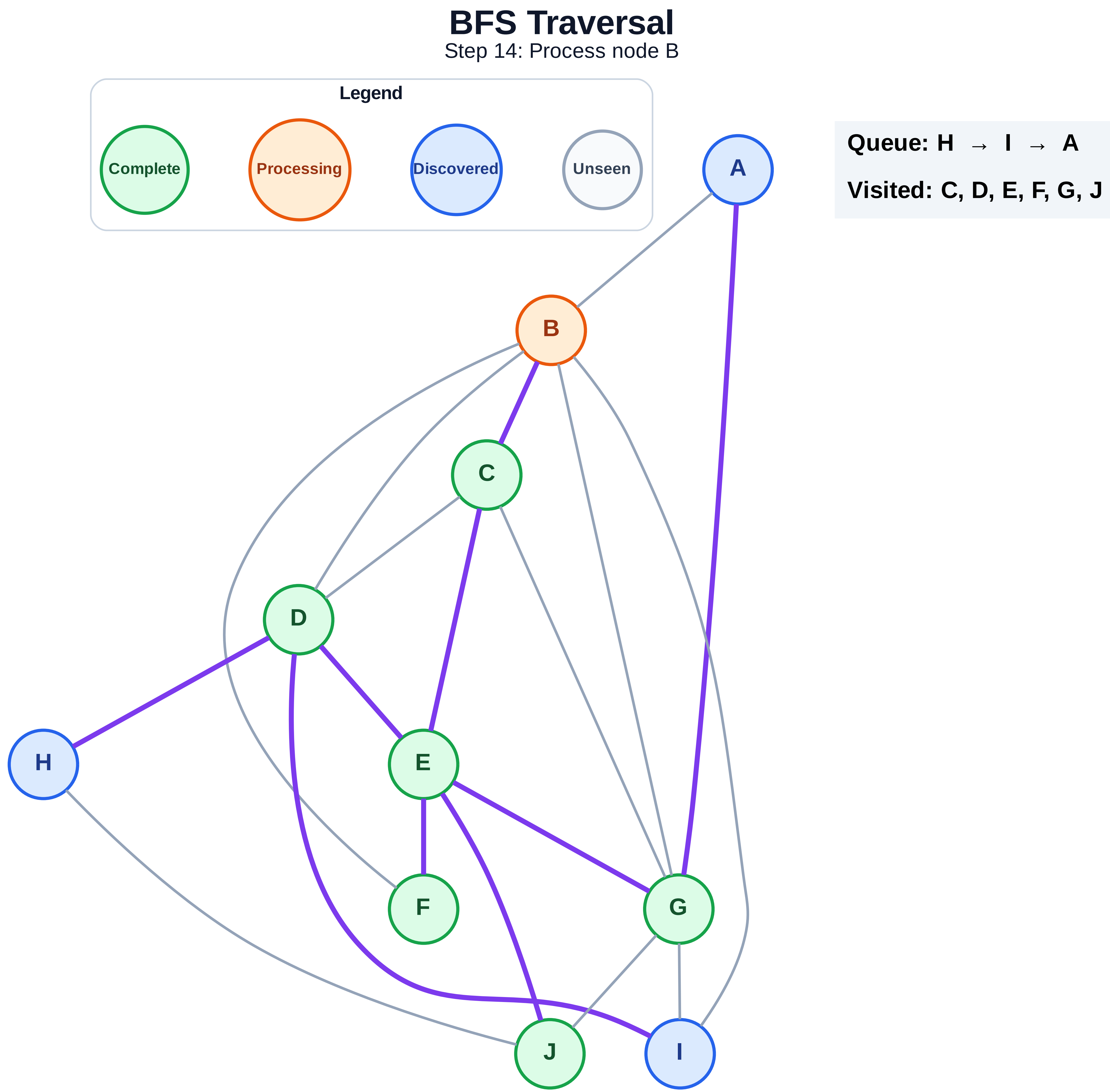
Step 12: Process node J



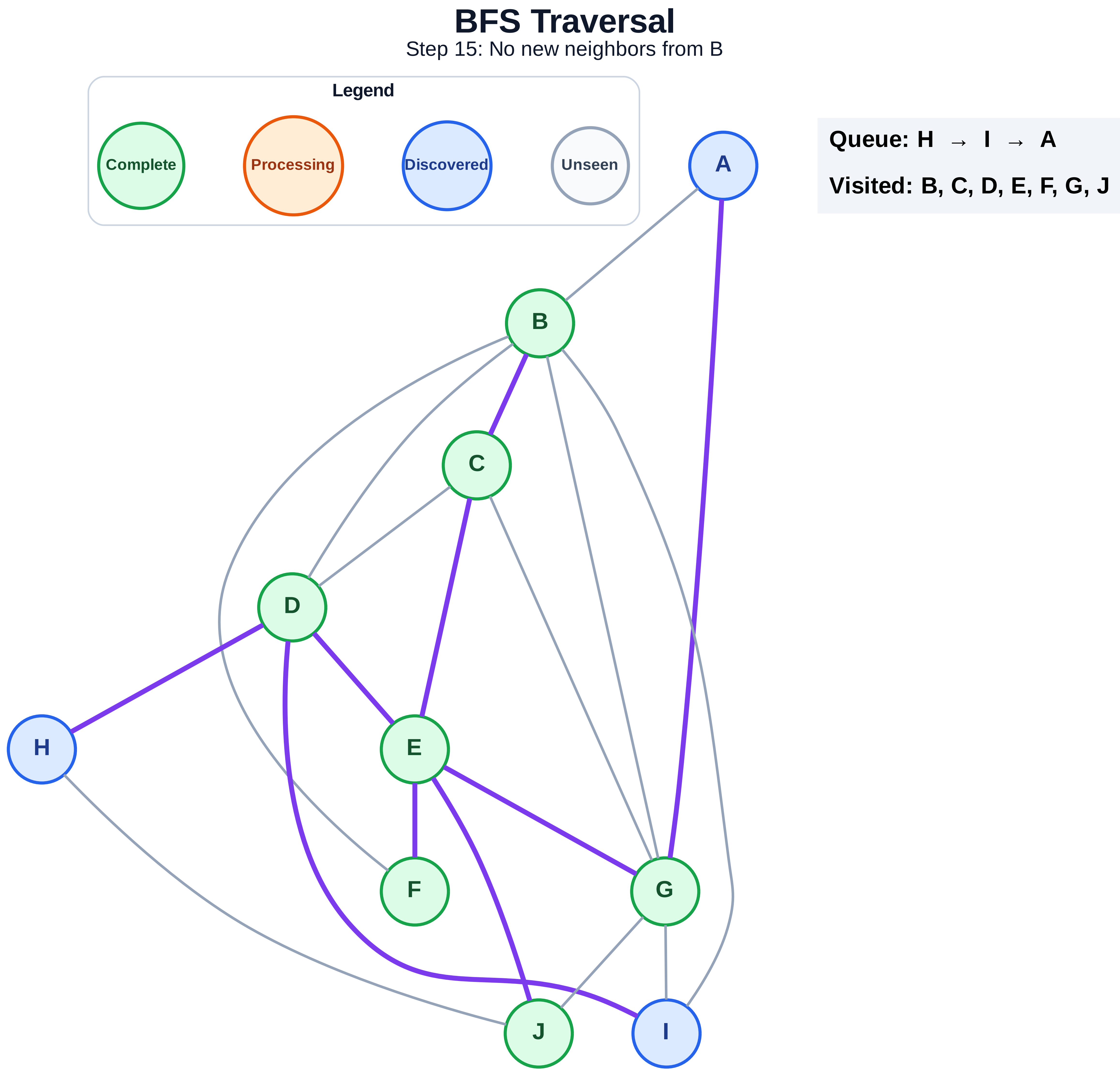
# BFS Traversal

Step 13: No new neighbors from J





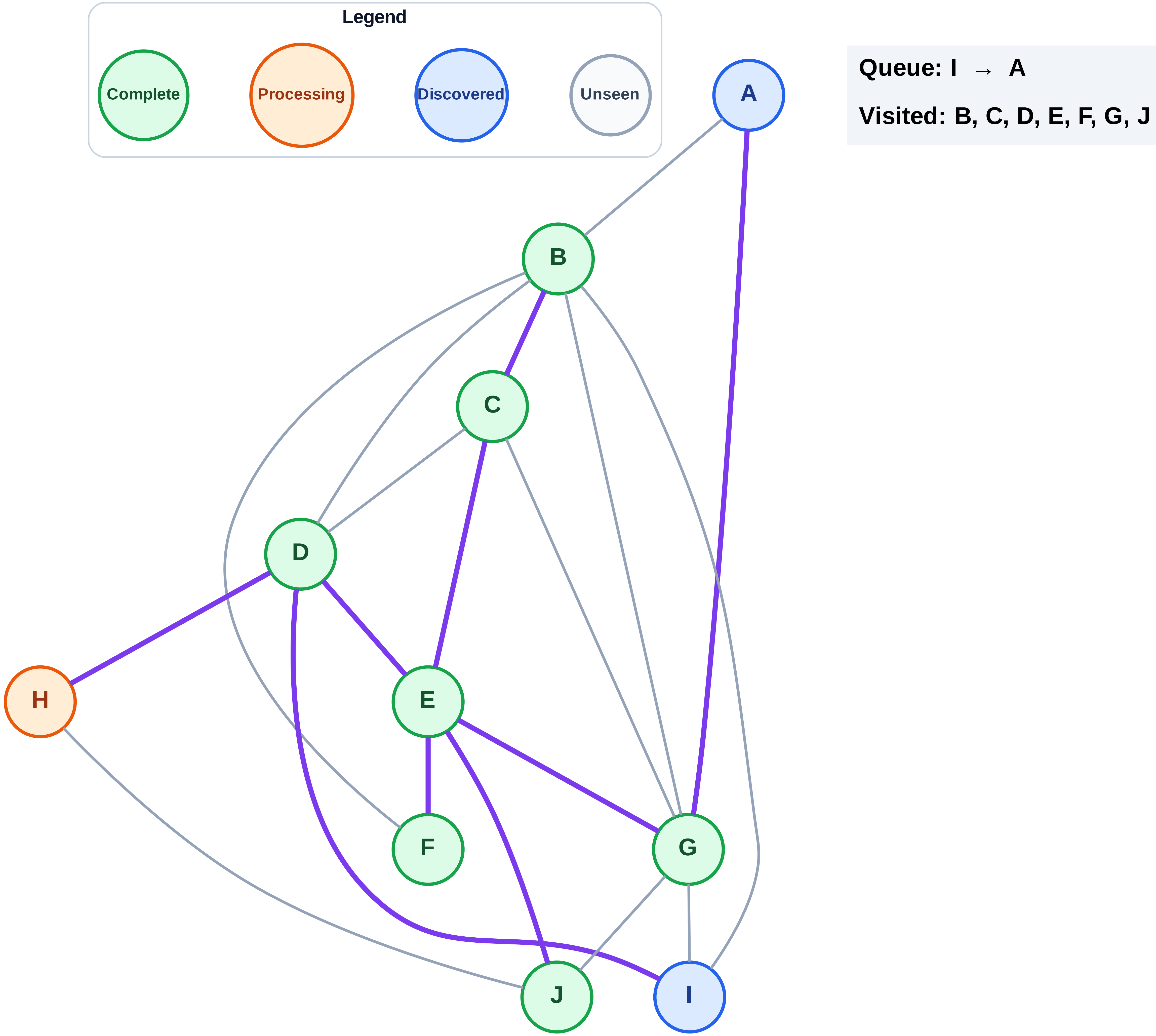
BFS Traversal
Step 14: Process node B
Legend
Complete
Processing
Discovered
Unseen
Queue: H → I → A
Visited: C, D, E, F, G, J
A
B
C
D
E
F
G
H
I
J

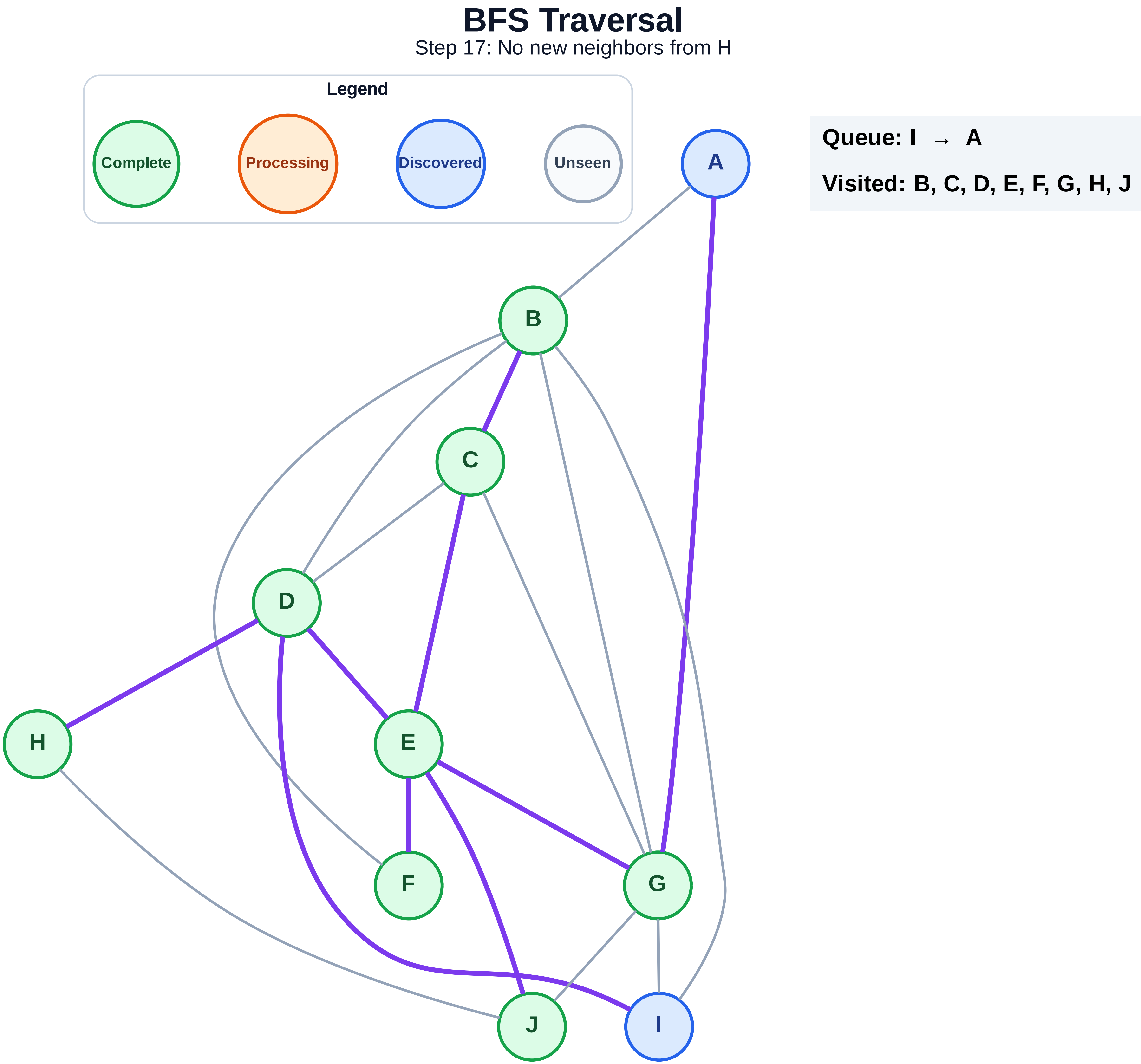


BFS Traversal
Step 15: No new neighbors from B
Legend
Complete
Processing
Discovered
Unseen
Queue: H → I → A
Visited: B, C, D, E, F, G, J
A
B
C
D
E
F
G
H
I
J

# BFS Traversal

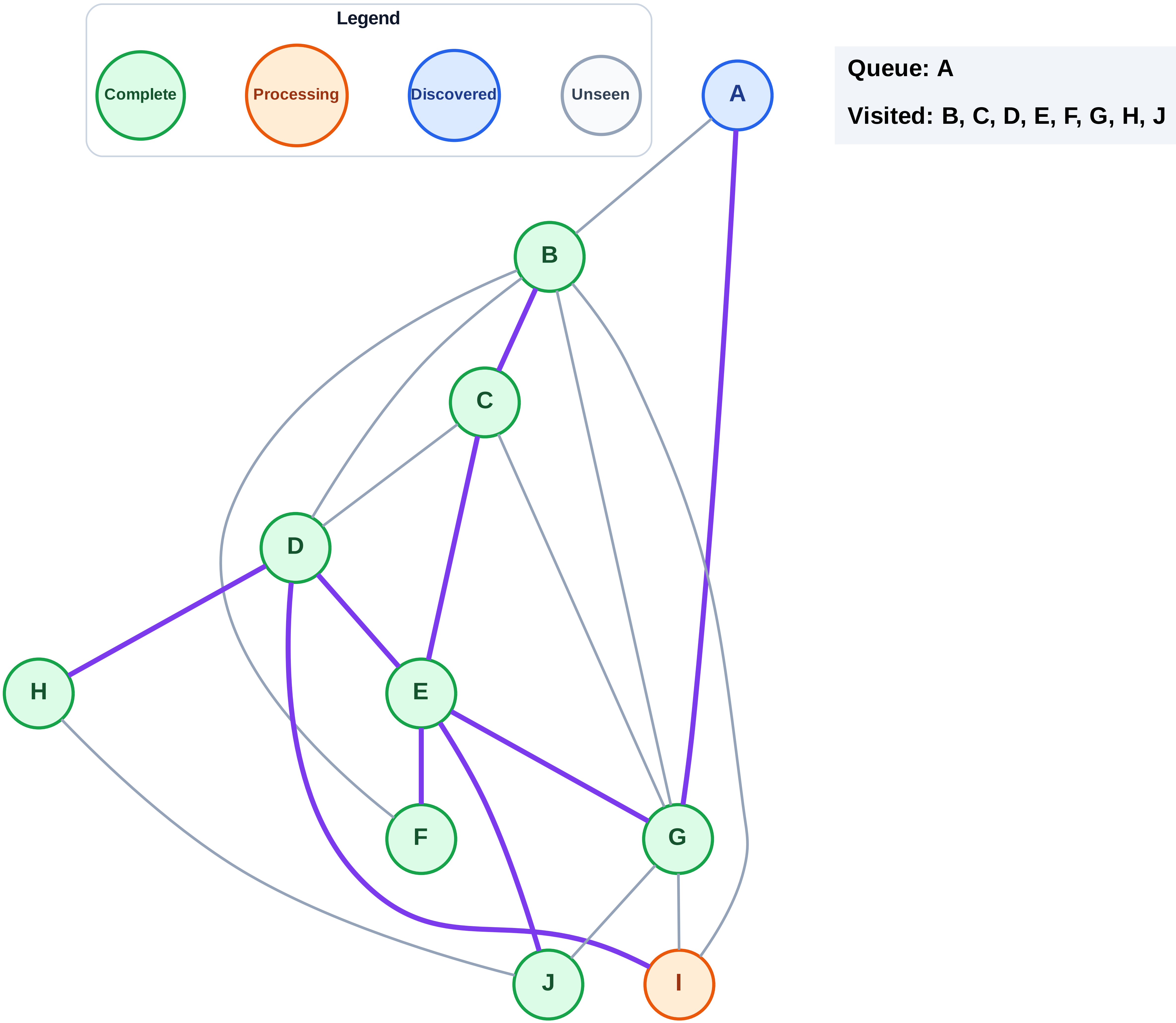
Step 16: Process node H

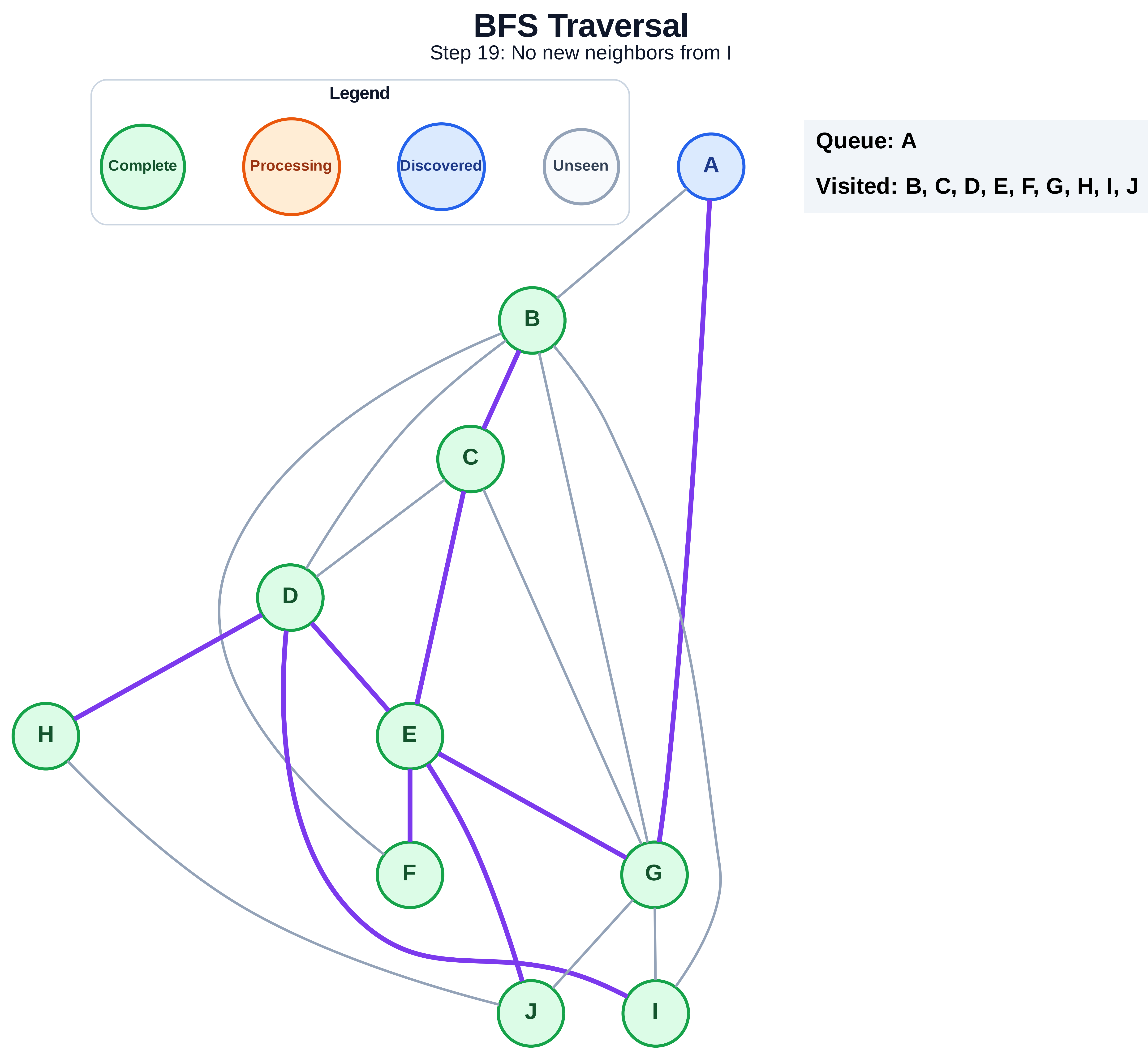



BFS Traversal
Step 17: No new neighbors from H
Legend
Complete
Processing
Discovered
Unseen
Queue: I → A
Visited: B, C, D, E, F, G, H, J
A
B
C
D
E
F
G
H
I
J

# BFS Traversal

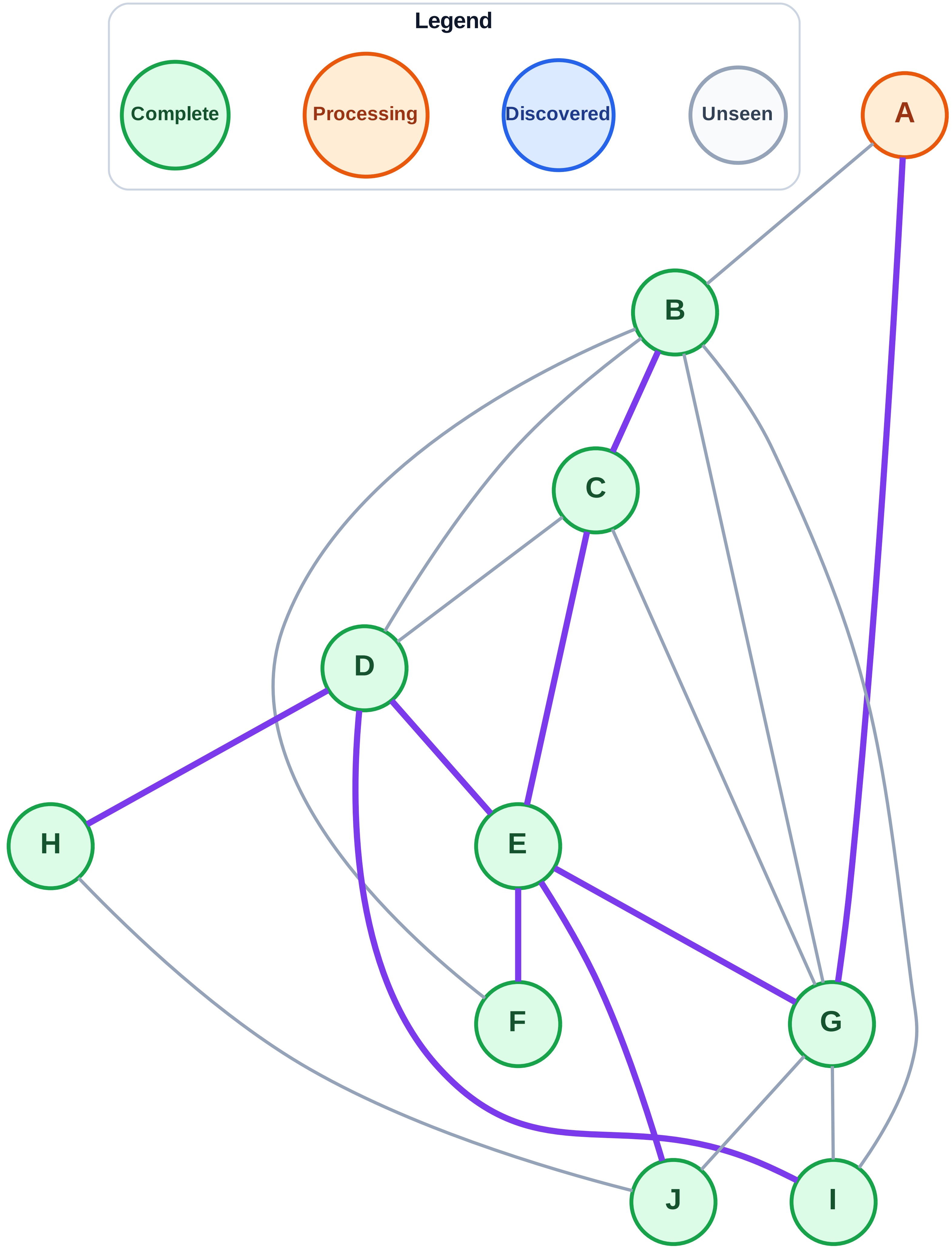
Step 18: Process node I




BFS Traversal
Step 19: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Queue: A
Visited: B, C, D, E, F, G, H, I, J
A
B
C
D
E
F
G
H
I
J

# BFS Traversal

Step 20: Process node A

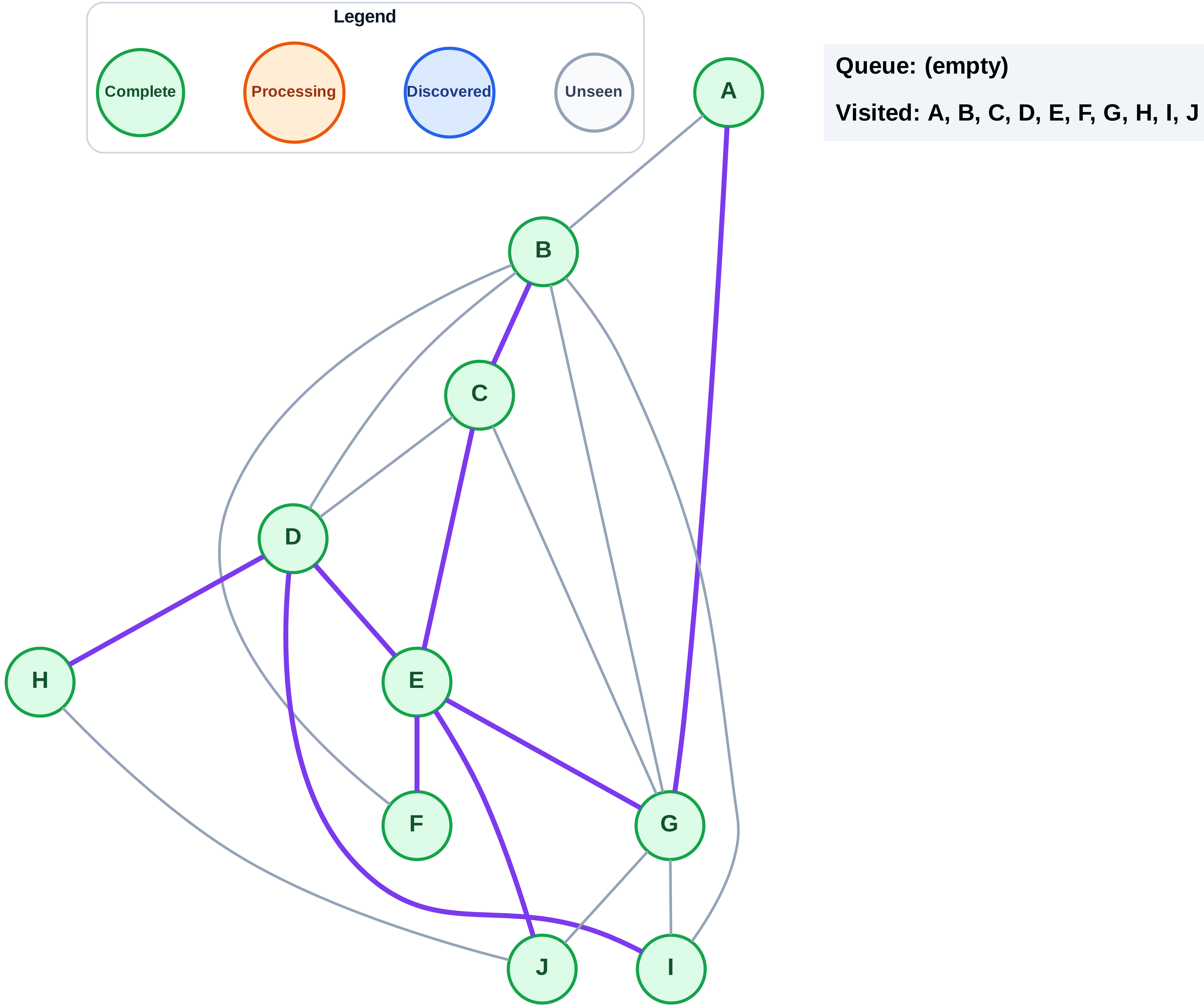



**Queue: (empty)**

**Visited: B, C, D, E, F, G, H, I, J**

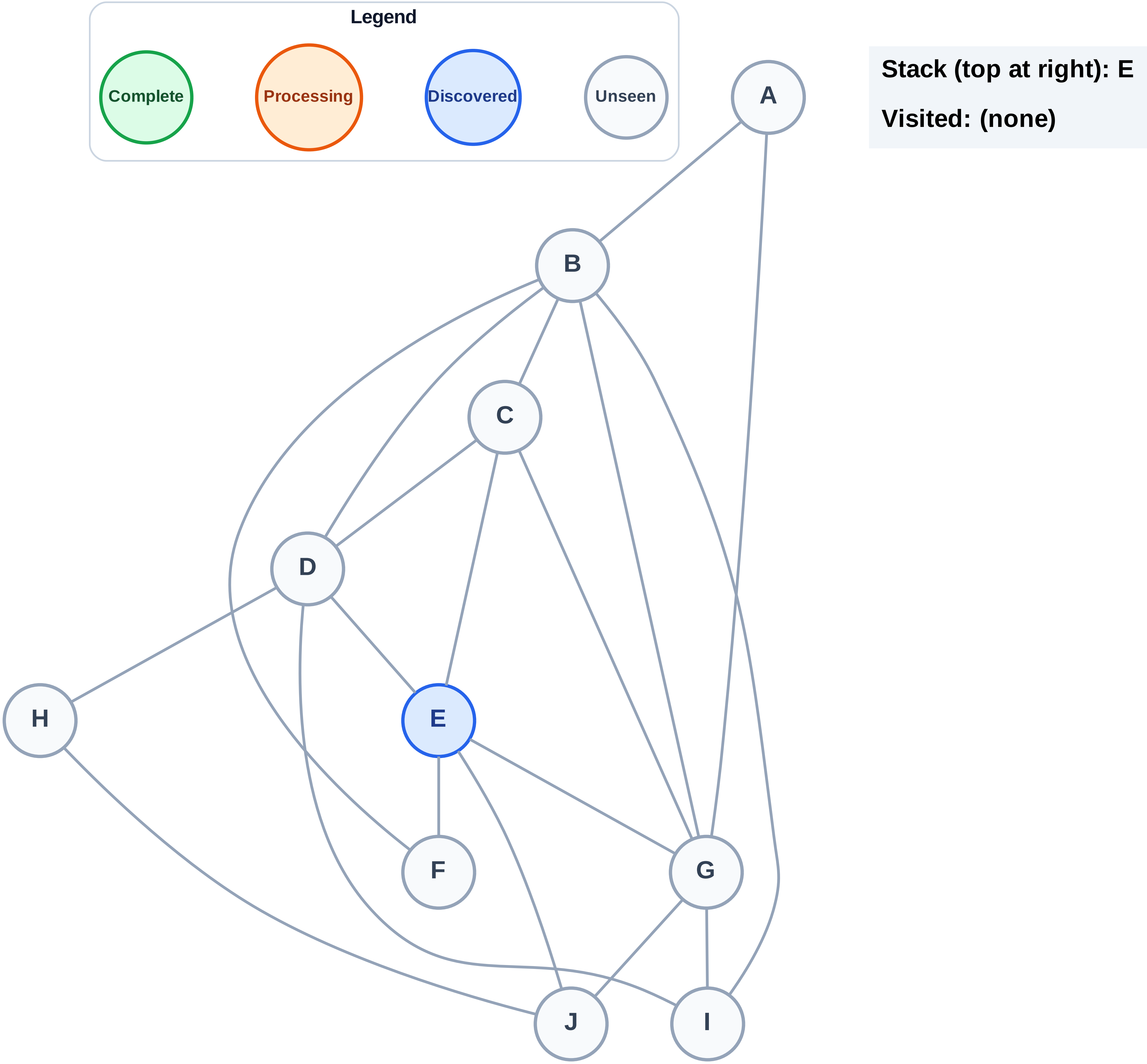
# BFS Traversal

Step 21: No new neighbors from A



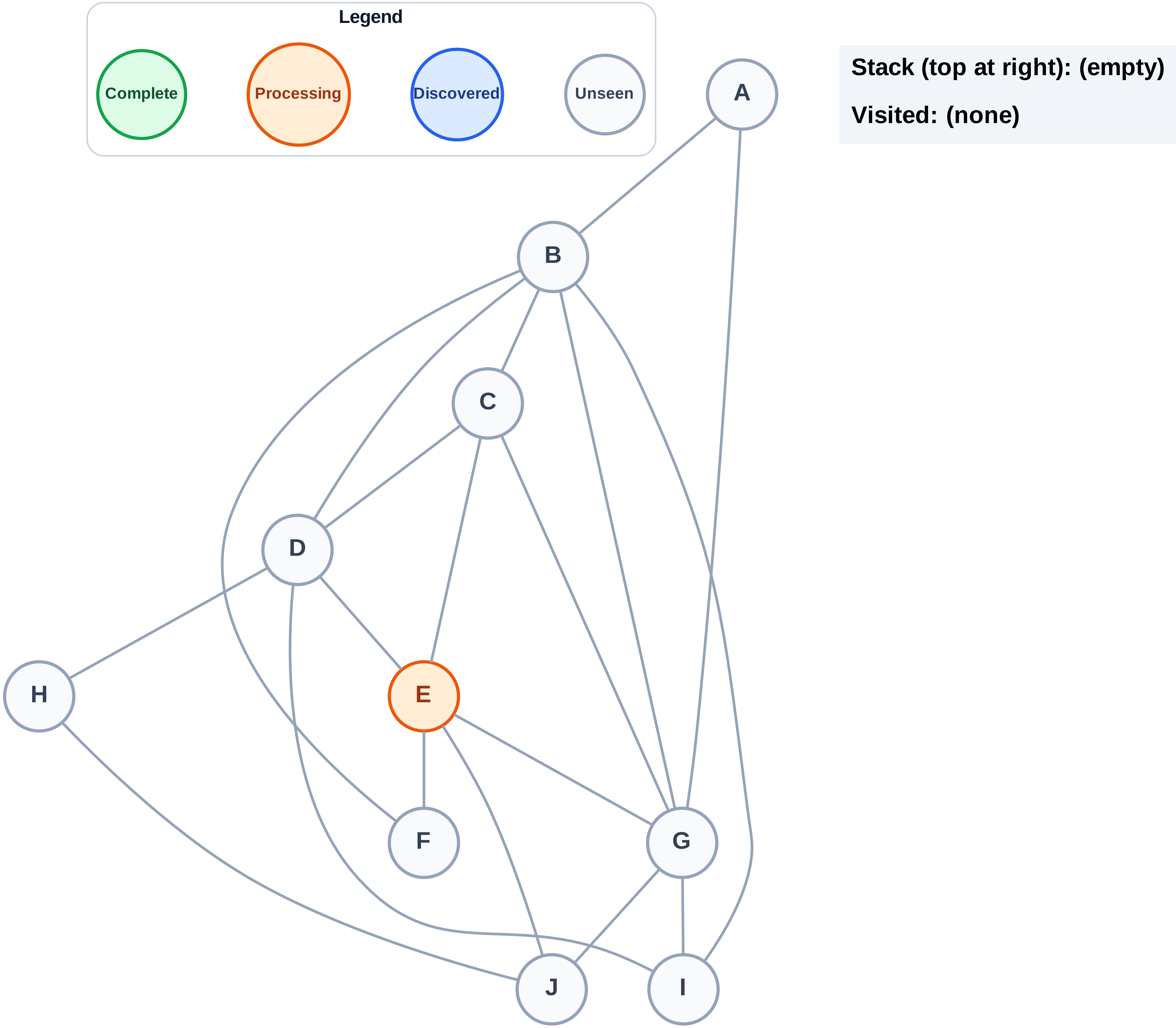
# DFS Traversal

Step 1: Start at E



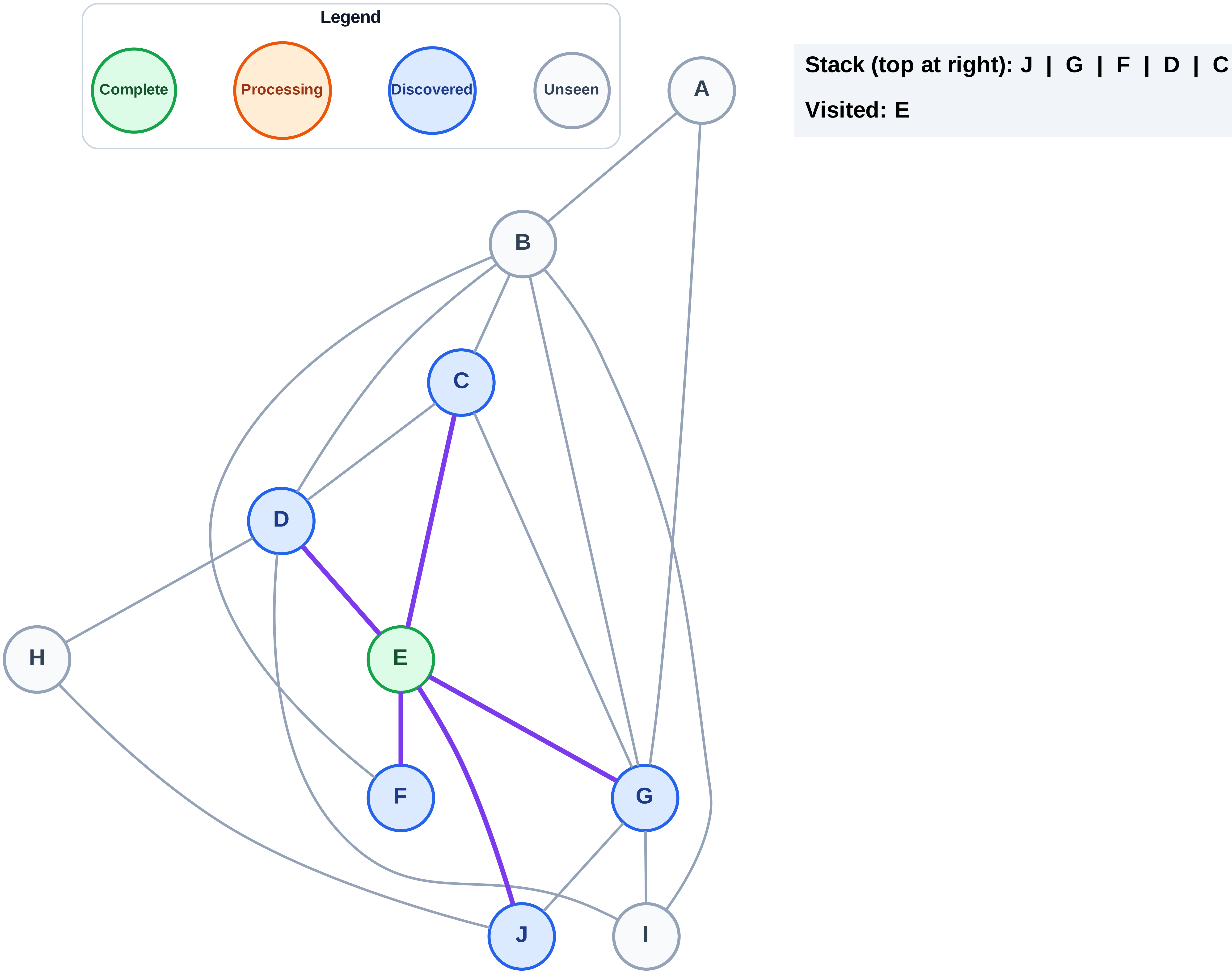
# DFS Traversal

Step 2: Process node E



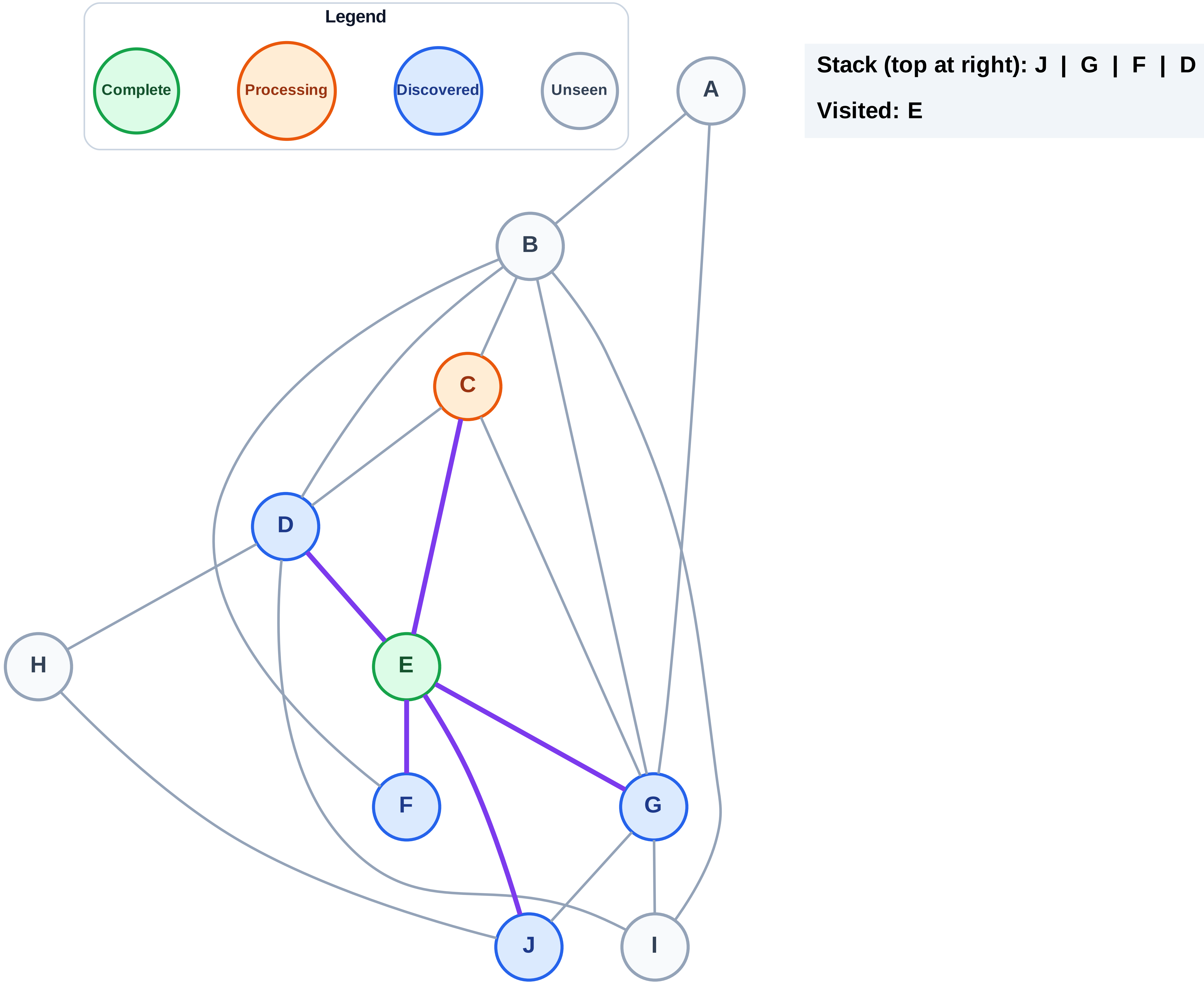
# DFS Traversal

Step 3: Discover J, G, F, D, C from E



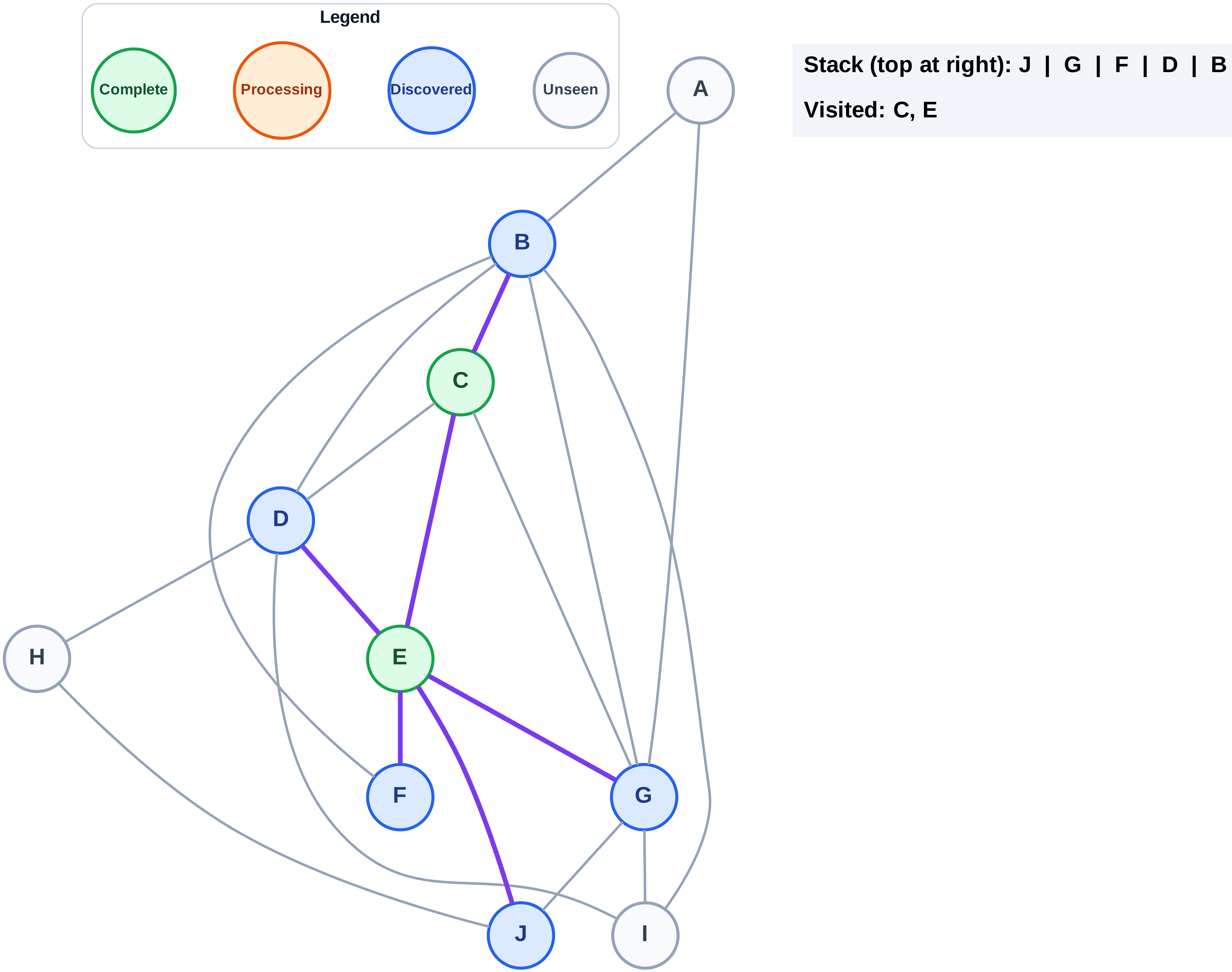
# DFS Traversal

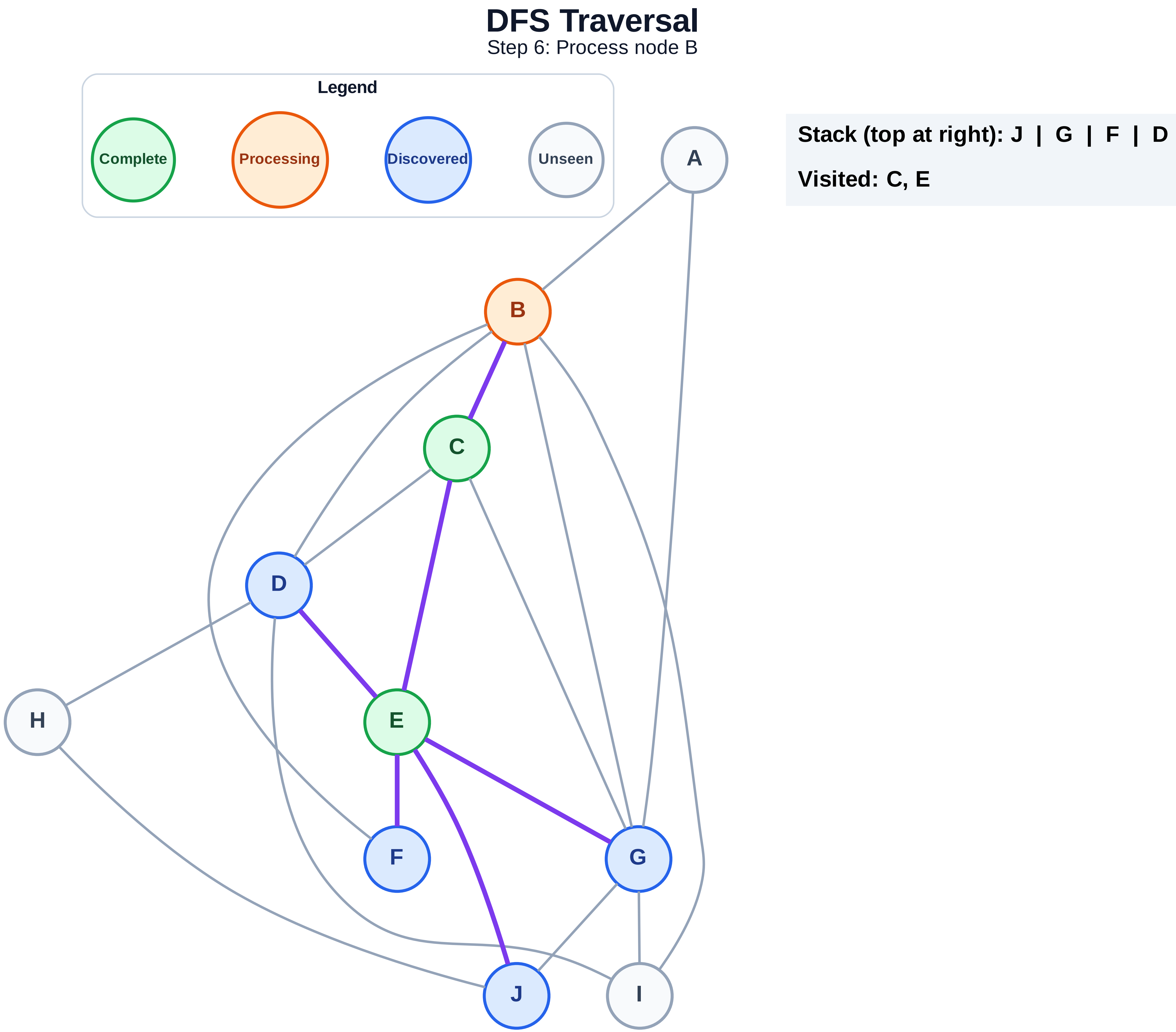
Step 4: Process node C



# DFS Traversal

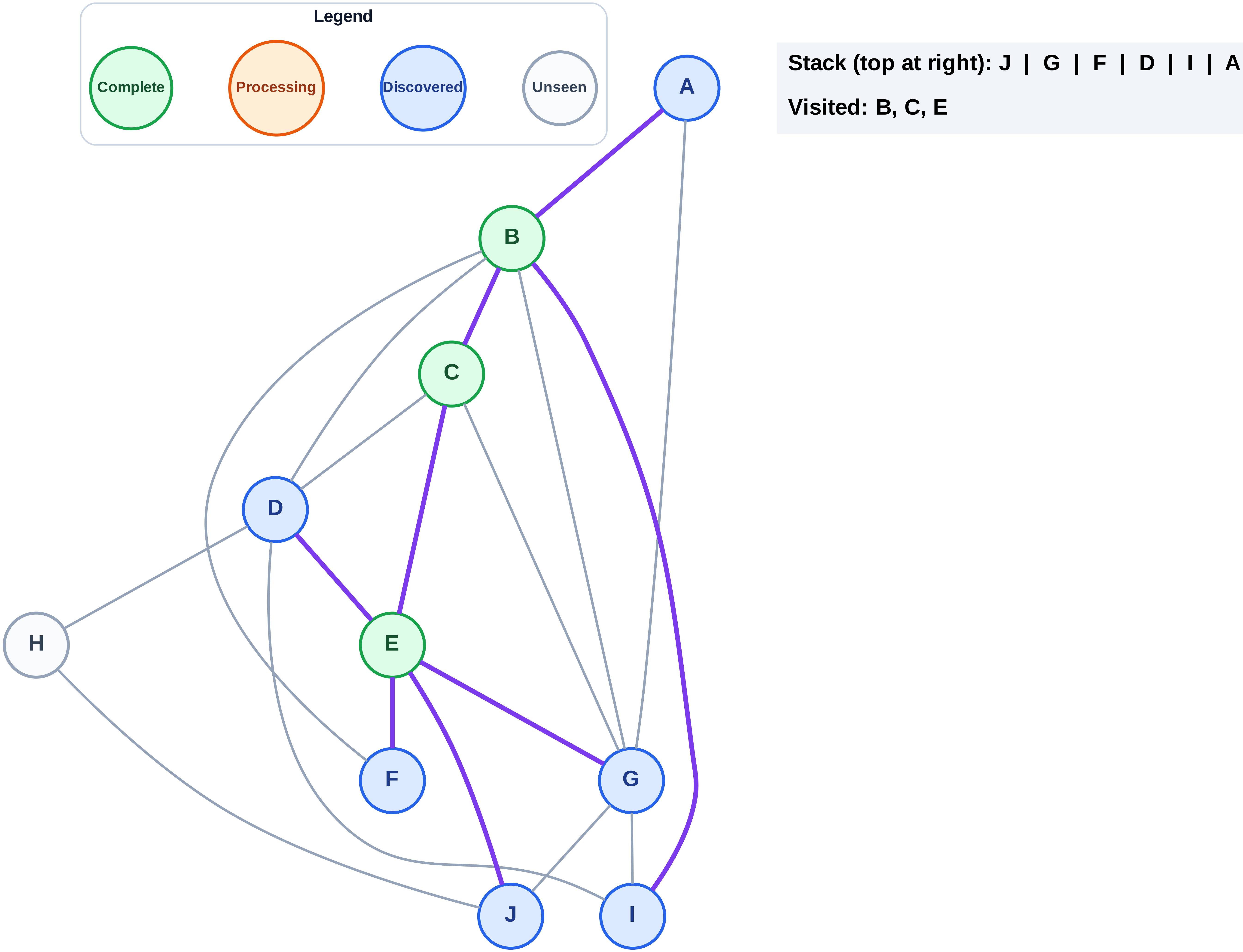
Step 5: Discover B from C




DFS Traversal
Step 6: Process node B
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | G | F | D
Visited: C, E
A
B
C
D
E
F
G
H
I
J

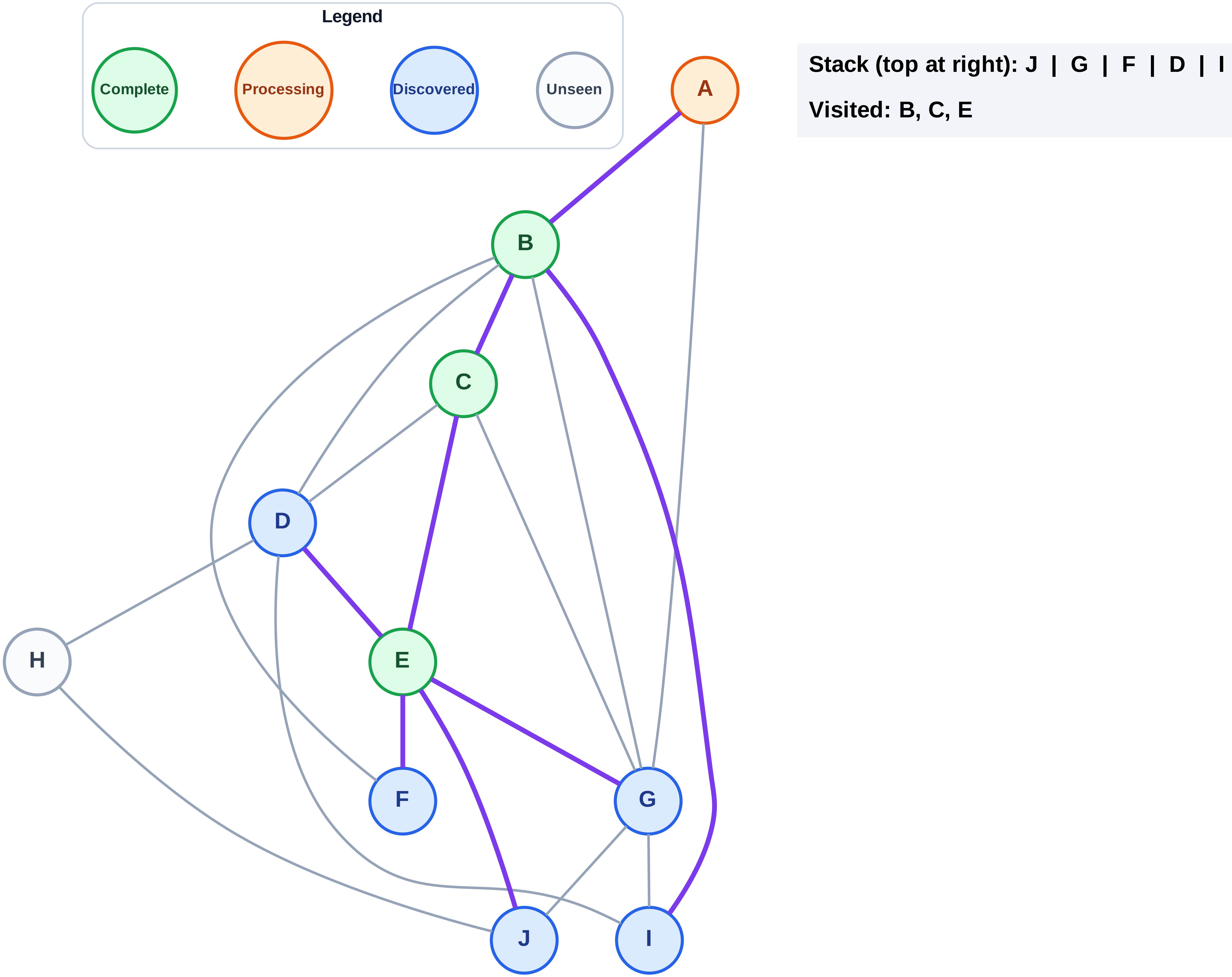
# DFS Traversal

Step 7: Discover I, A from B



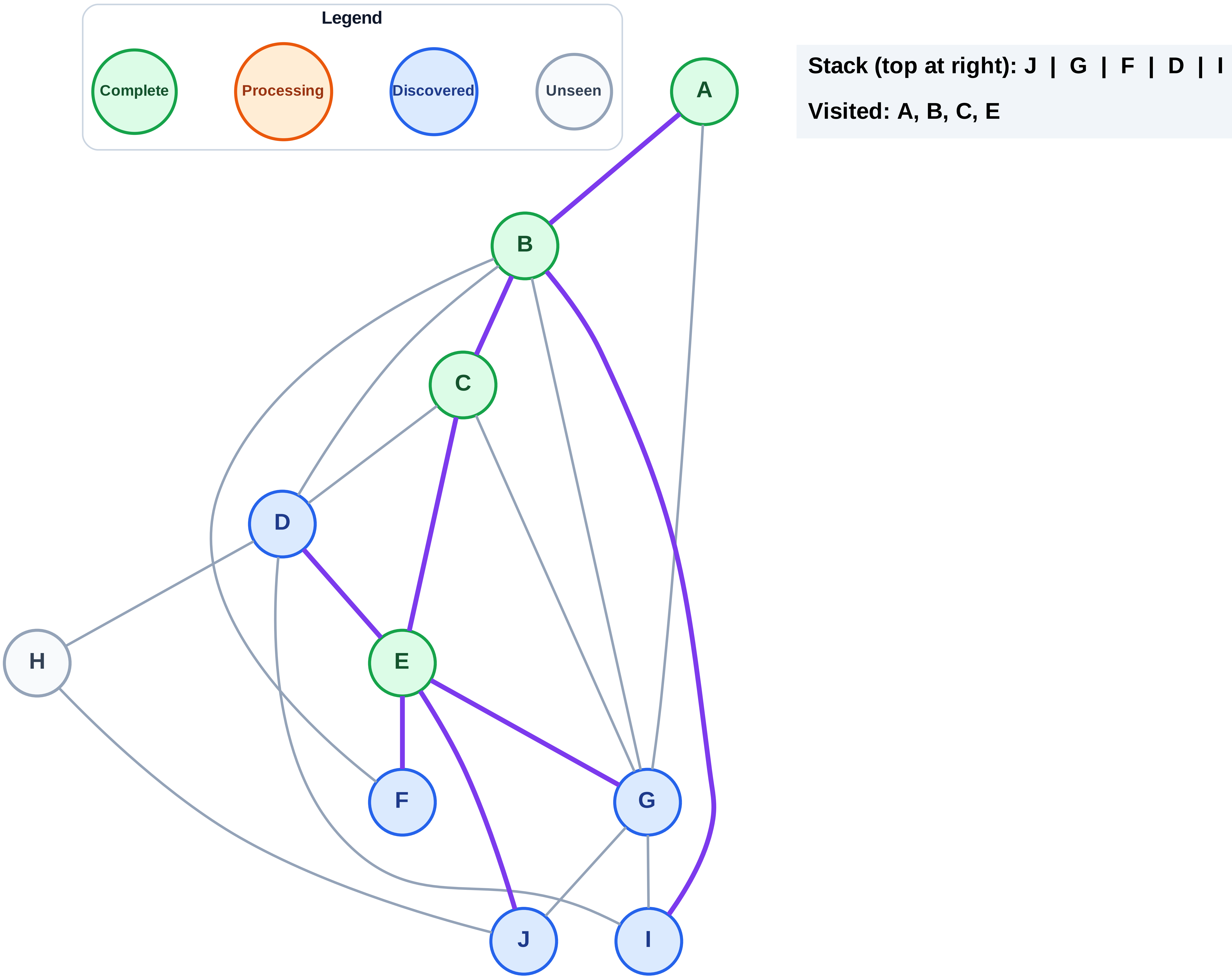
# DFS Traversal

Step 8: Process node A



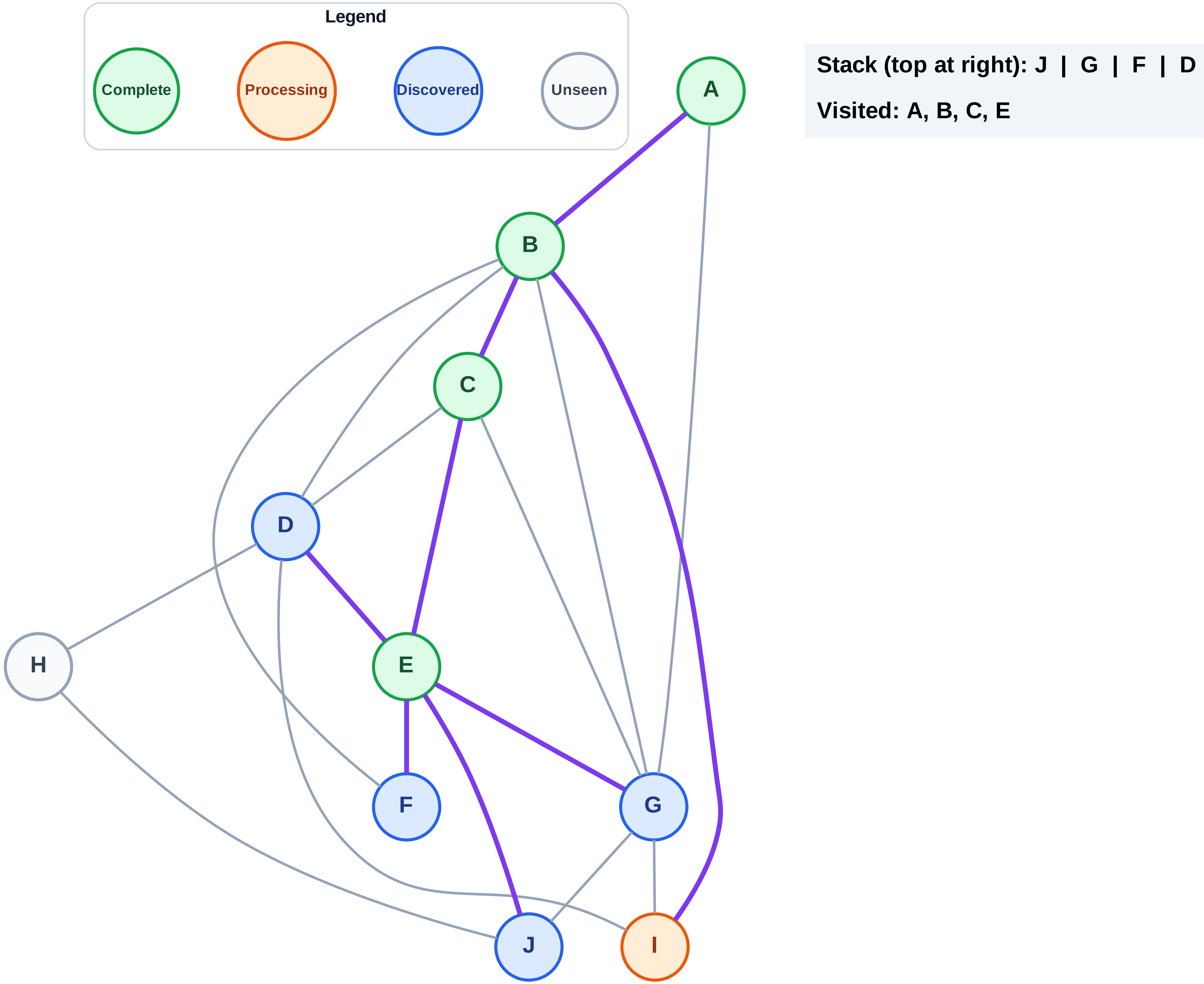
# DFS Traversal

Step 9: No new neighbors from A



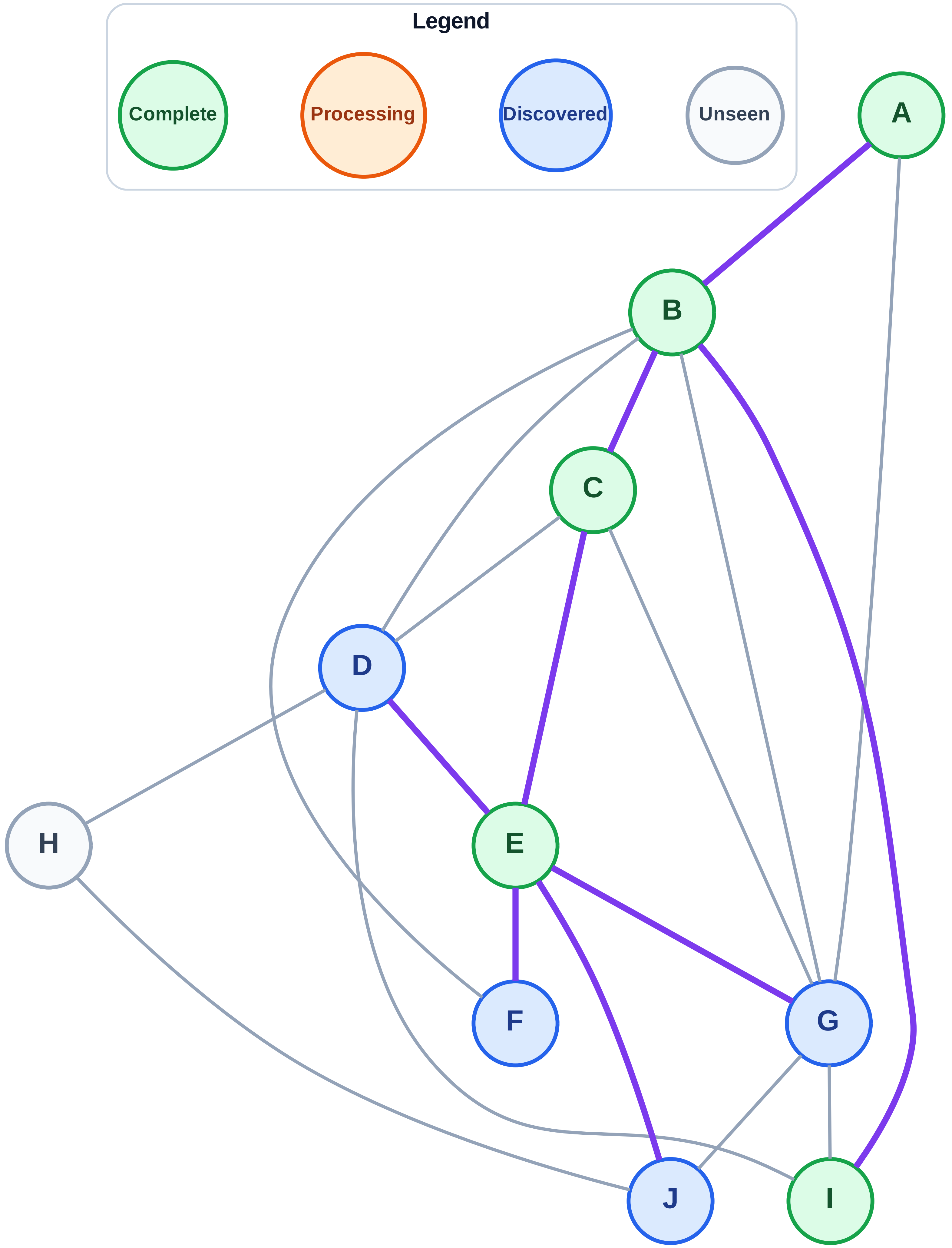
# DFS Traversal

Step 10: Process node I



# DFS Traversal

Step 11: No new neighbors from I

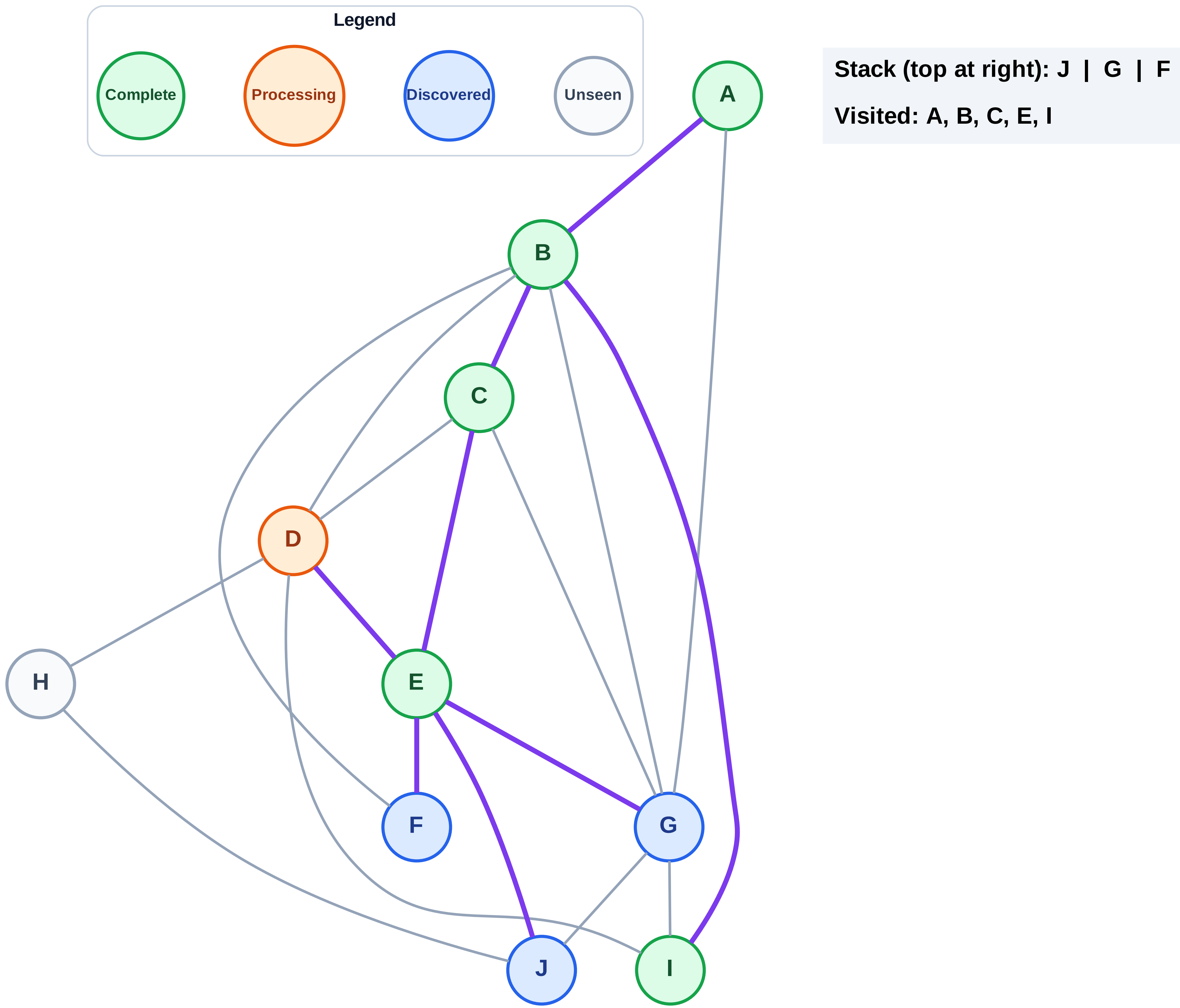



**Stack (top at right): J | G | F | D**

**Visited: A, B, C, E, I**

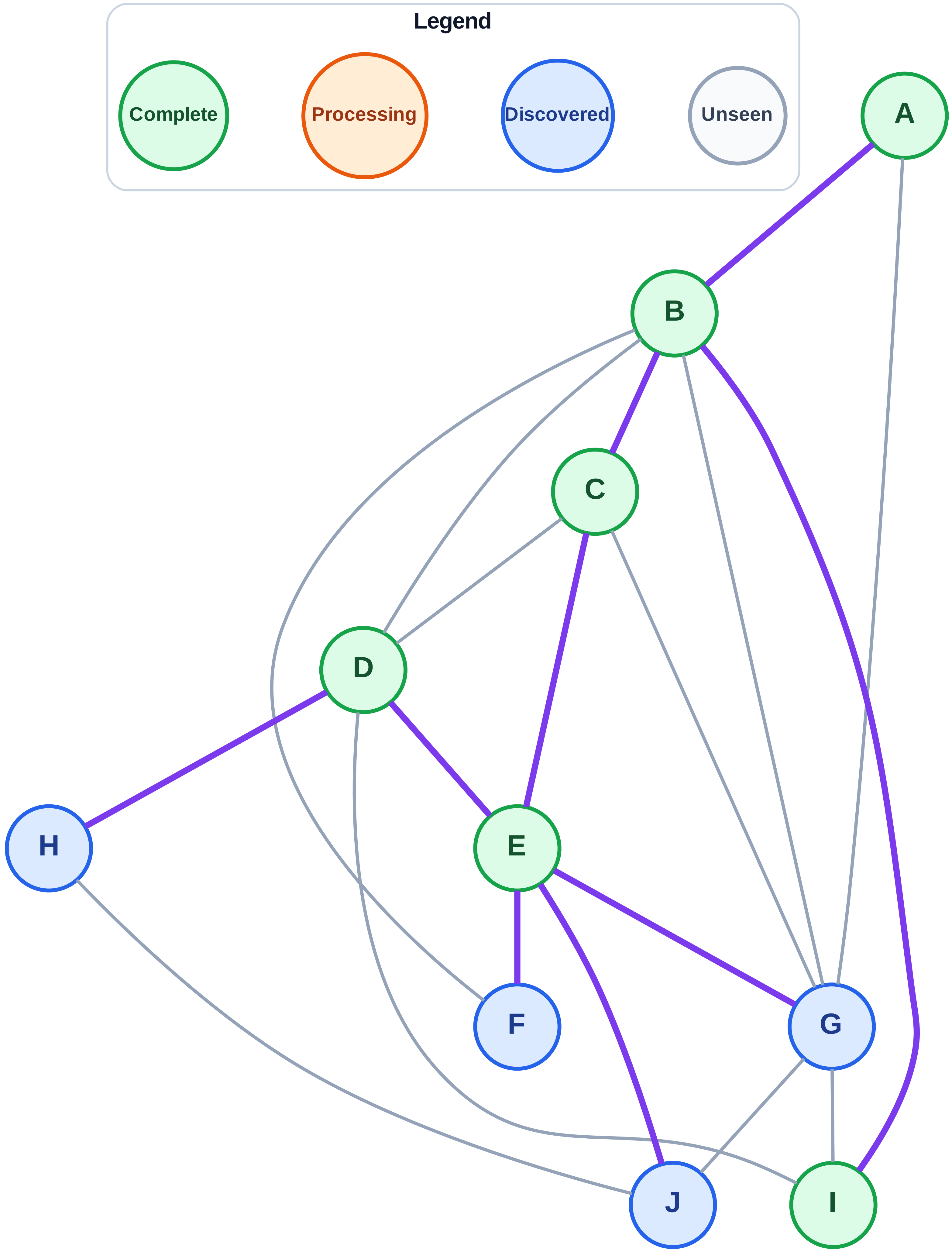
# DFS Traversal

Step 12: Process node D



# DFS Traversal

Step 13: Discover H from D

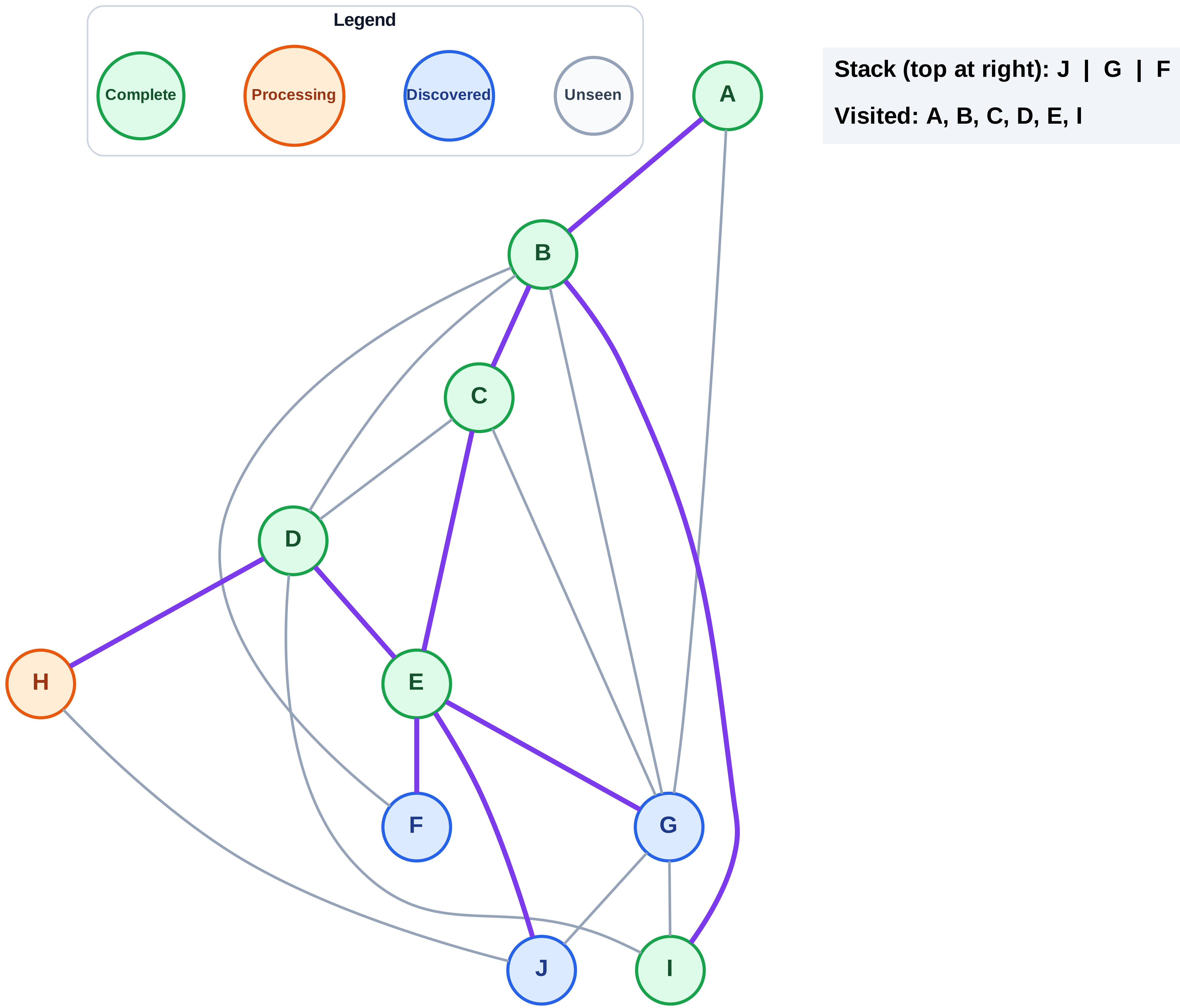



**Stack (top at right): J | G | F | H**

**Visited: A, B, C, D, E, I**

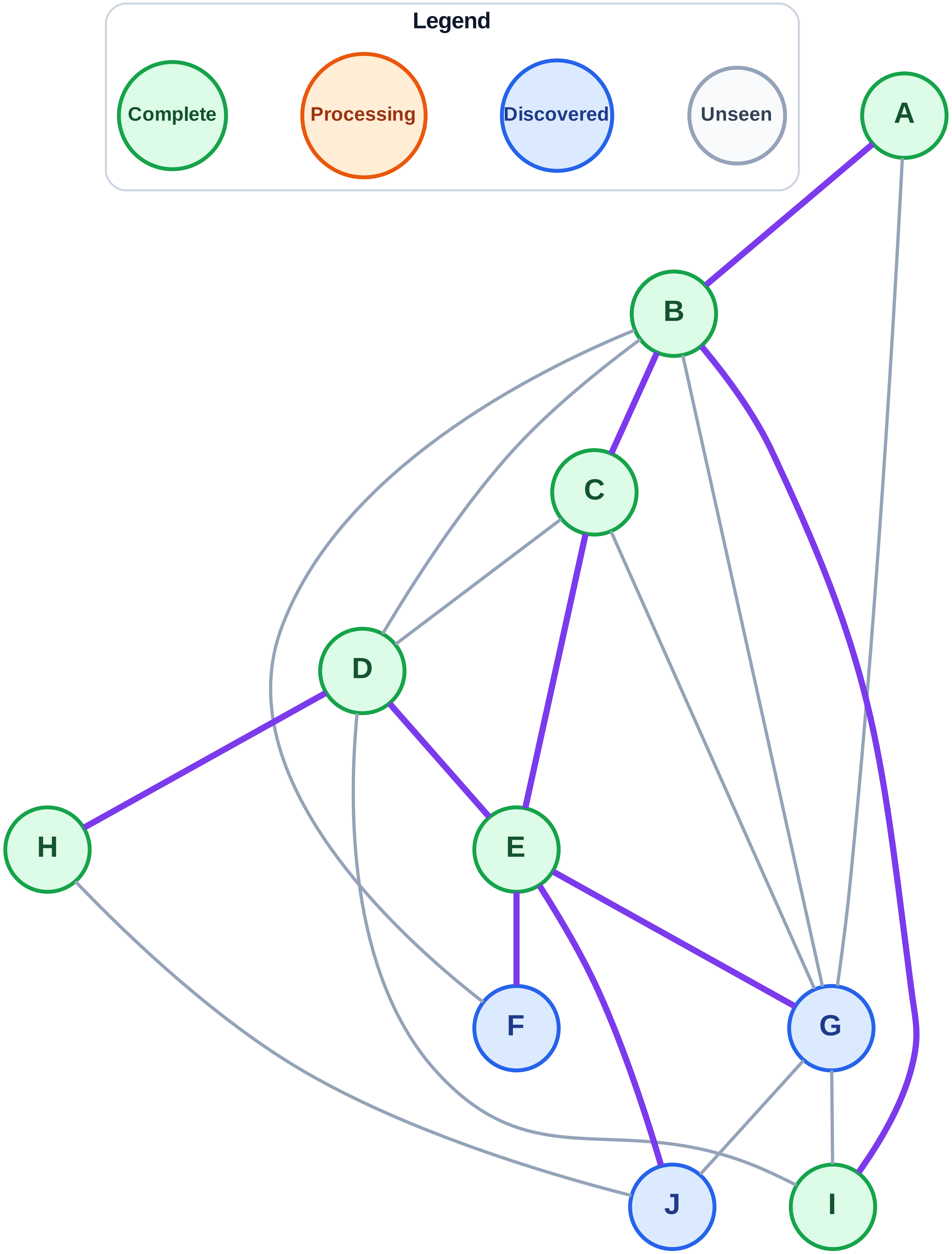
# DFS Traversal

Step 14: Process node H



# DFS Traversal

Step 15: No new neighbors from H

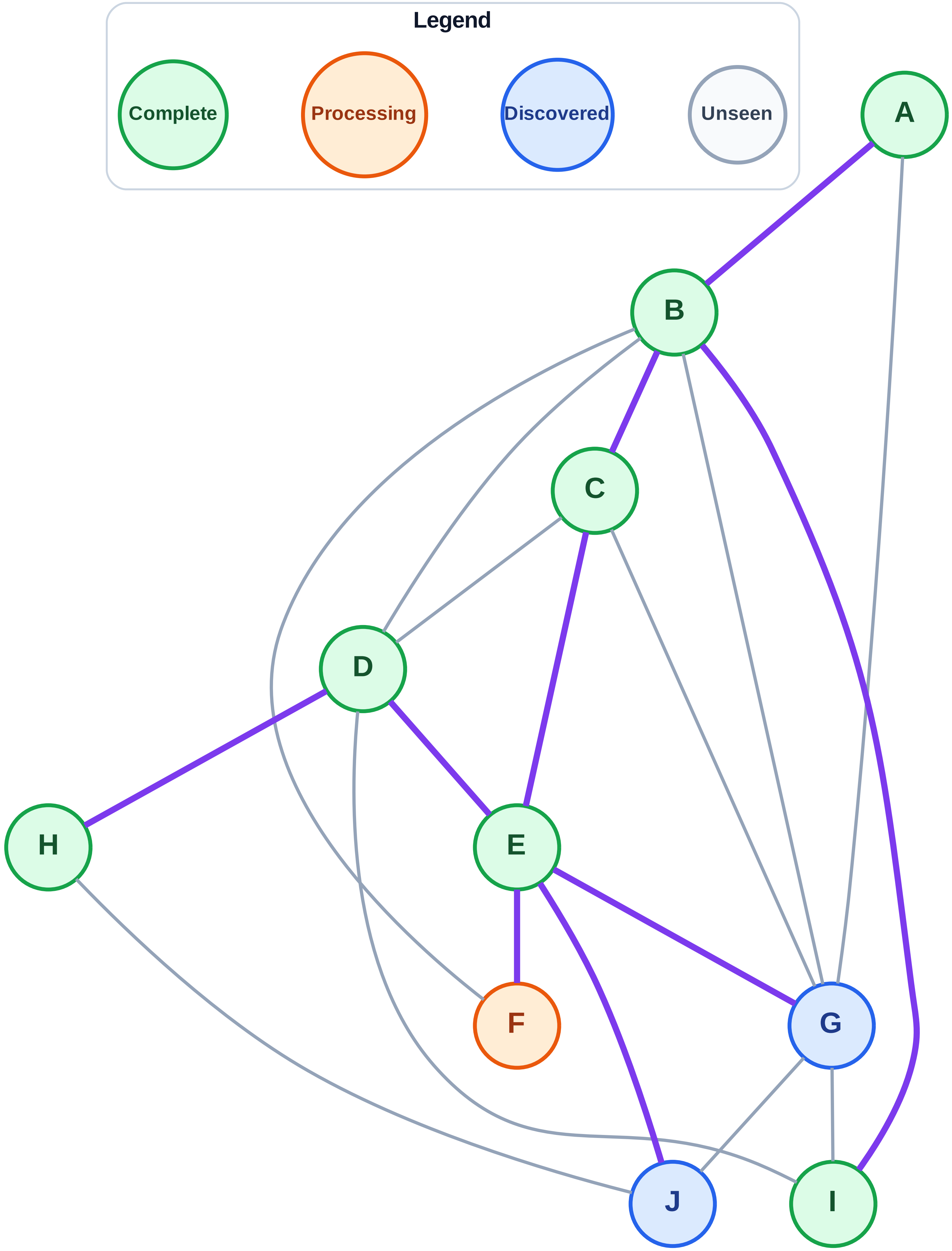



**Stack (top at right): J | G | F**

**Visited: A, B, C, D, E, H, I**

# DFS Traversal

Step 16: Process node F

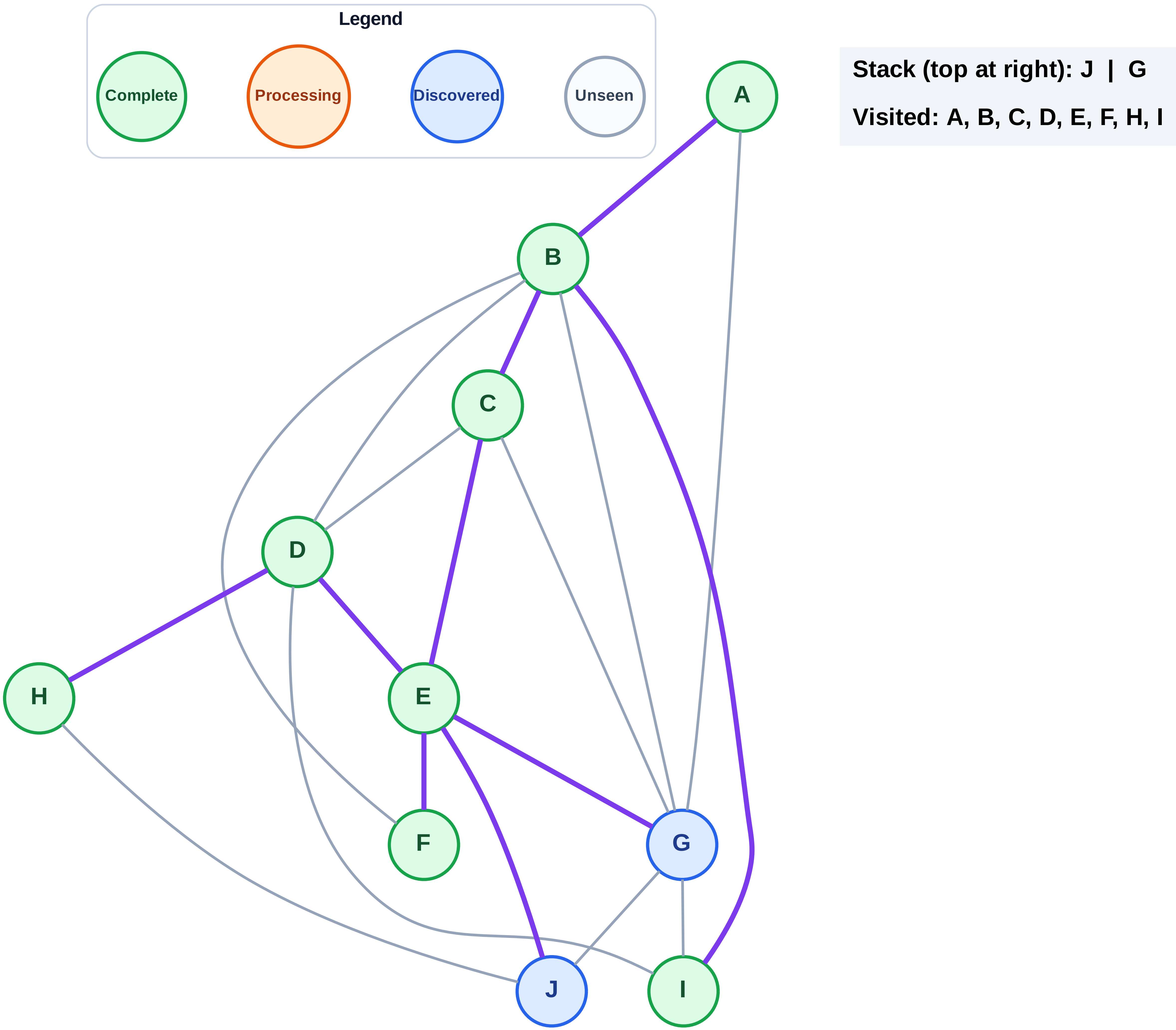


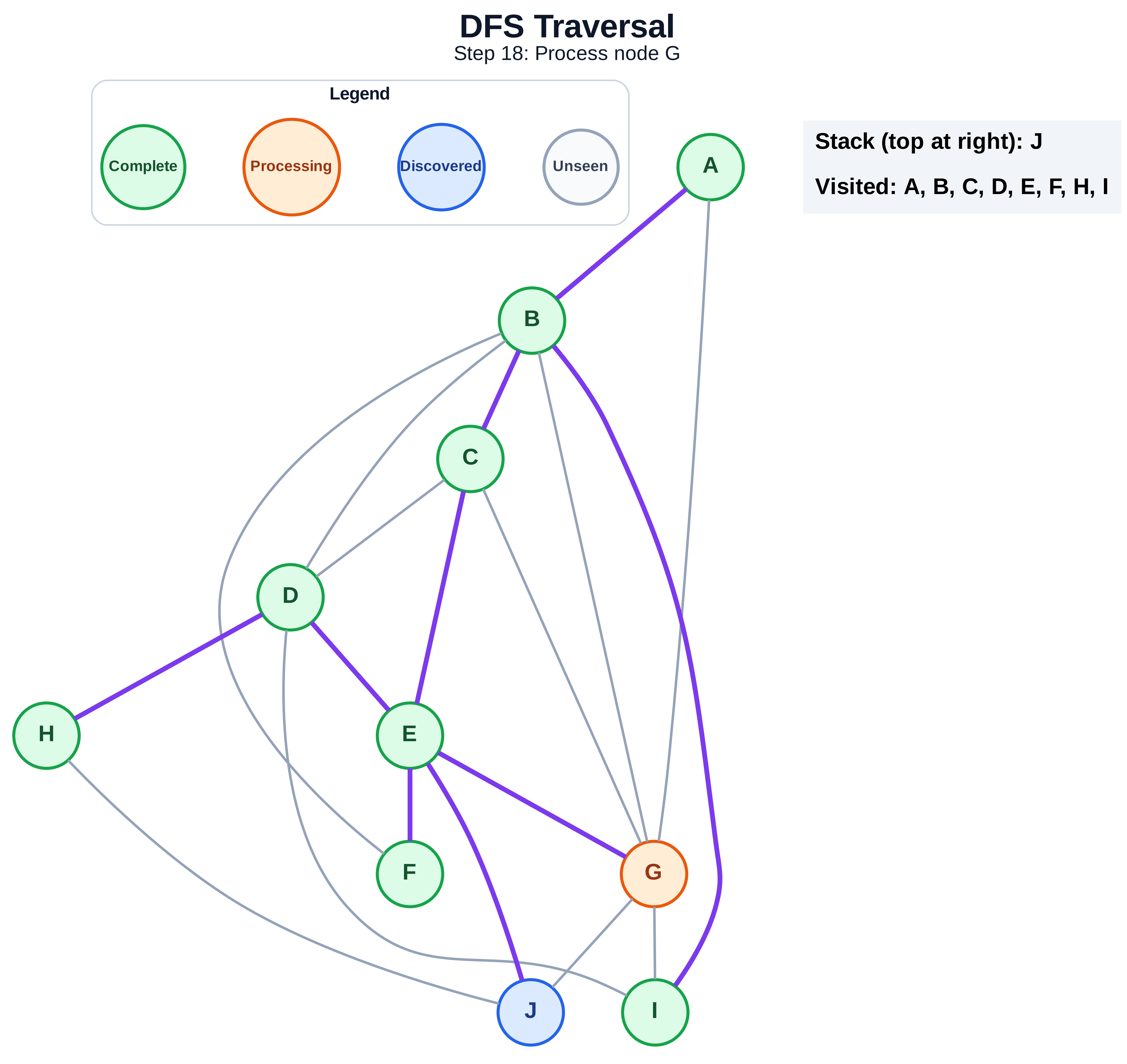

**Stack (top at right): J | G**

**Visited: A, B, C, D, E, H, I**

# DFS Traversal

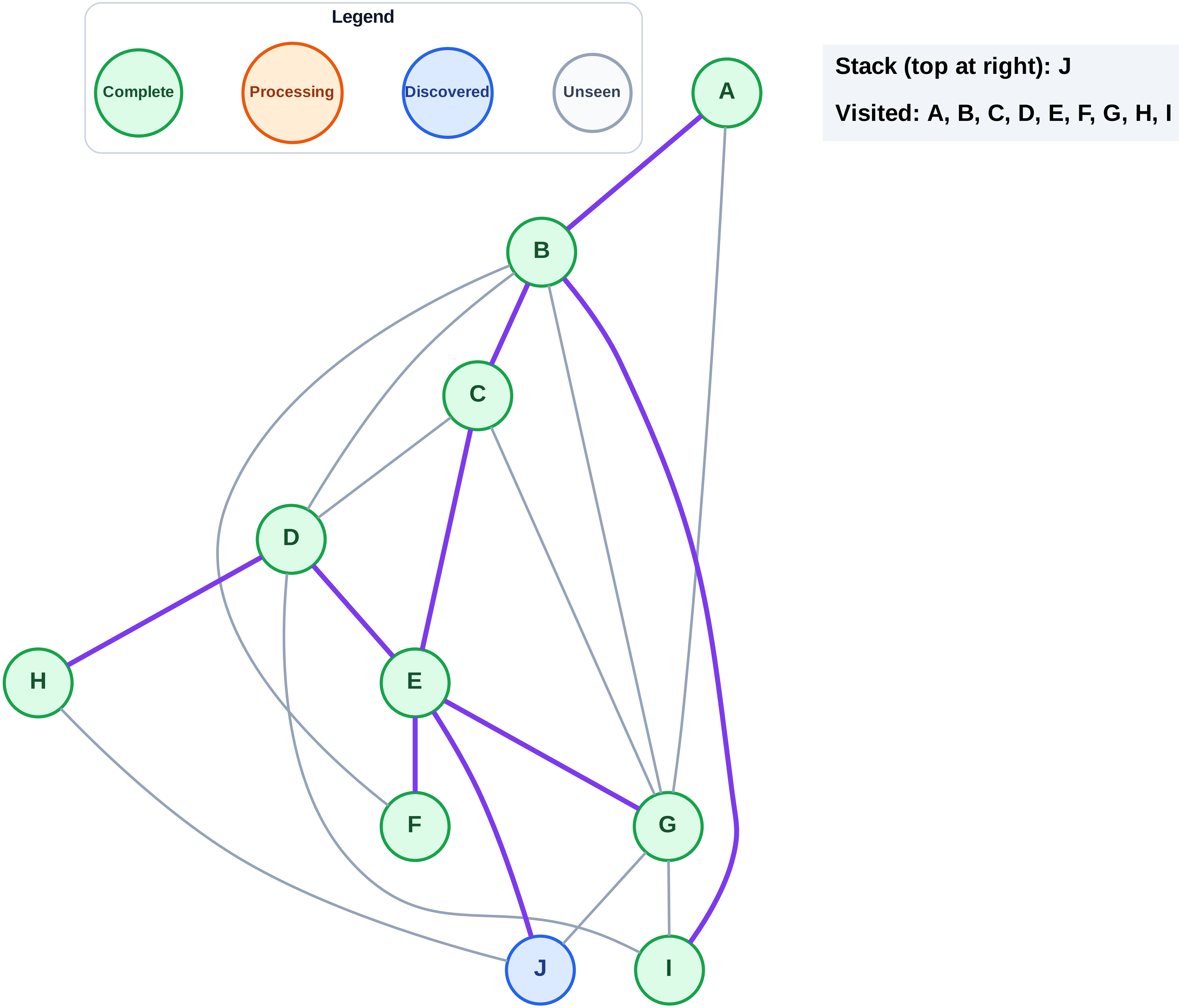
Step 17: No new neighbors from F




DFS Traversal
Step 18: Process node G
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J
Visited: A, B, C, D, E, F, H, I
A
B
C
D
E
F
G
H
I
J

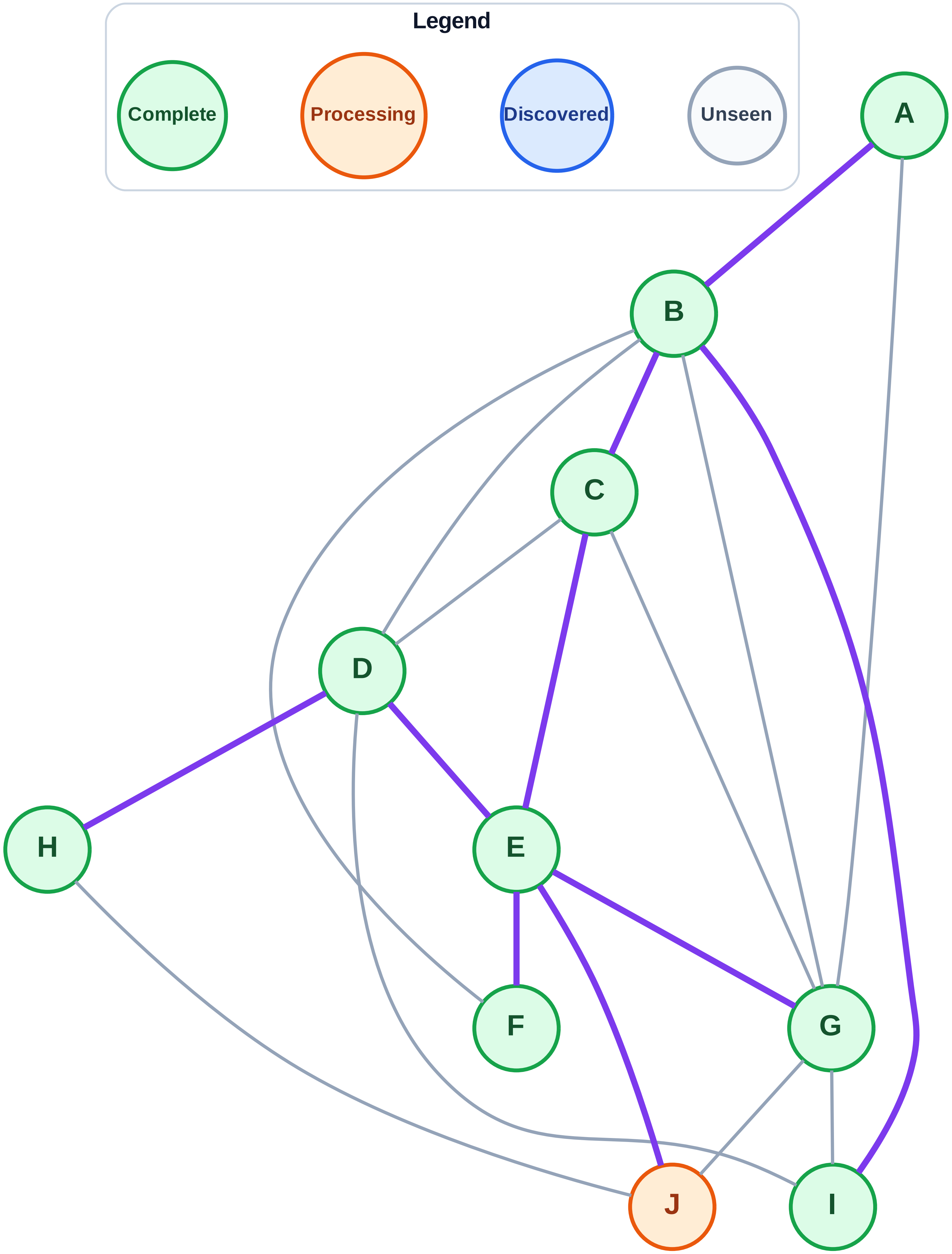
# DFS Traversal

Step 19: No new neighbors from G



# DFS Traversal

Step 20: Process node J




**Stack (top at right): (empty)**

**Visited: A, B, C, D, E, F, G, H, I**

# DFS Traversal

Step 21: No new neighbors from J